

Linear Momentum and Collisions

- ❑ Conservation of Energy
- ❑ Momentum
- ❑ Impulse
- ❑ Conservation of Momentum
- ❑ 1-D Collisions
- ❑ 2-D Collisions
- ❑ The Center of Mass



Linear Momentum

- ❑ This is a new fundamental quantity, like force, energy. It is a vector quantity (points in same direction as velocity).
- ❑ The linear momentum **p** of an object of mass m moving with a velocity **v** is defined to be the product of the mass and velocity:

$$\vec{p} = m \vec{v}$$

- ❑ The terms momentum and linear momentum will be used interchangeably in the text
- ❑ Momentum depend on an object's mass and velocity



Linear Momentum, cont'd

- Linear momentum is a vector quantity $\vec{\mathbf{p}} = m\vec{\mathbf{v}}$
 - Its direction is the same as the direction of the velocity
- The dimensions of momentum are ML/T
- The SI units of momentum are kg m / s
- Momentum can be expressed in component form:

$$p_x = mv_x \quad p_y = mv_y \quad p_z = mv_z$$



Newton's Law and Momentum

- Newton's Second Law can be used to relate the momentum of an object to the resultant force acting on it

$$\vec{F}_{net} = m \vec{a} = m \frac{\Delta \vec{v}}{\Delta t} = \frac{\Delta (m \vec{v})}{\Delta t}$$

- The change in an object's momentum divided by the elapsed time equals the constant net force acting on the object

$$\frac{\Delta \vec{p}}{\Delta t} = \frac{\text{change in momentum}}{\text{time interval}} = \vec{F}_{net}$$



Impulse

- When a single, constant force acts on the object, there is an **impulse** delivered to the object

$$\vec{I} = \vec{F} \Delta t$$

-
- $\vec{I}$ is defined as the *impulse*
- The equality is true even if the force is not constant
- Vector quantity, the direction is the same as the direction of the force

$$\frac{\Delta \vec{p}}{\Delta t} = \frac{\text{change in momentum}}{\text{time interval}} = \vec{F}_{net}$$

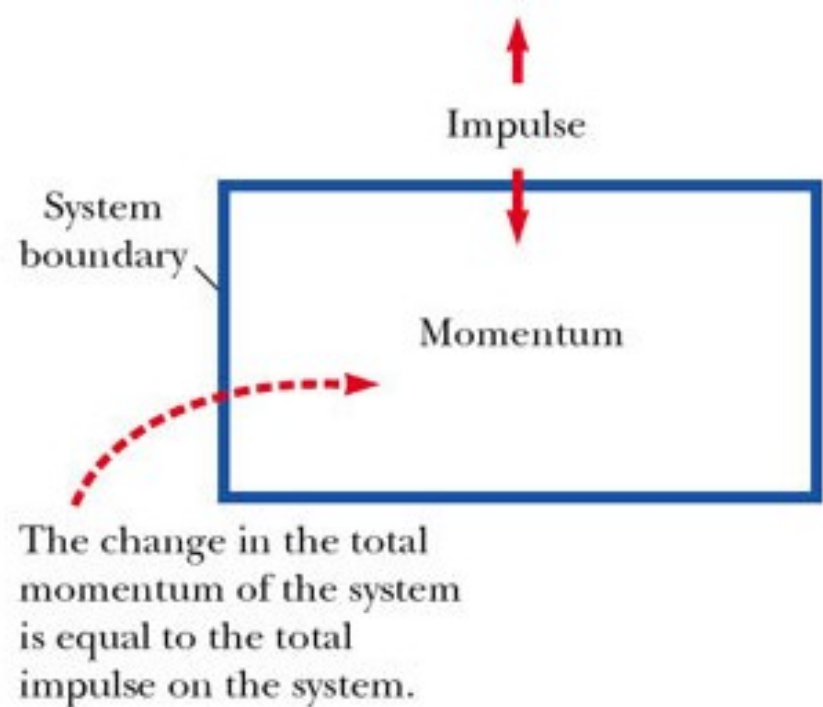


Impulse-Momentum Theorem

- The theorem states that the impulse acting on a system is equal to the change in momentum of the system

$$\Delta \vec{p} = \vec{F}_{net} \Delta t = \vec{I}$$

$$\vec{I} = \Delta \vec{p} = m \vec{v}_f - m \vec{v}_i$$

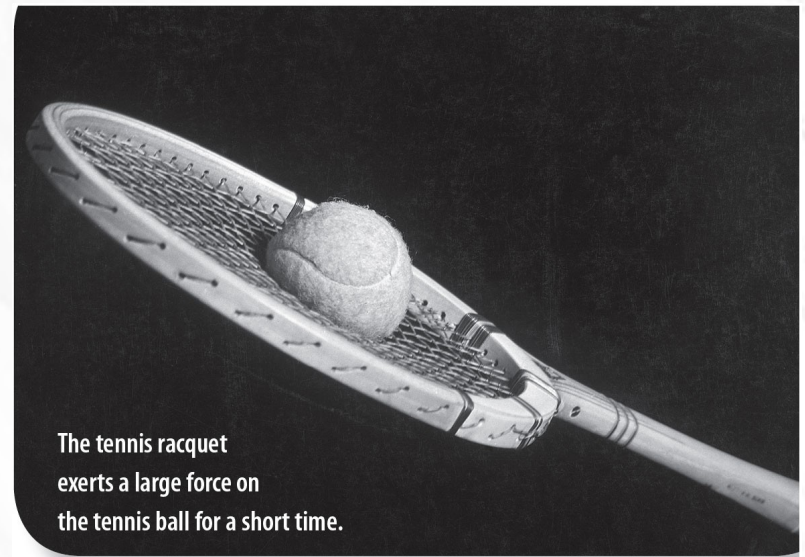


$$F\Delta t = m\Delta V$$

Is a different way to write Newton's second law → apply a force during a time t and there is a change in momentum. Or if there is a change in momentum, there is a force applied during a time t (bouncing ball on a wall).

Let's estimate the average force on a tennis ball as it is served.

The ball's mass is 0.06 kg and it leaves the racket with a speed of 40 m/s. High-speed photography indicates that the contact time is about 5 milliseconds.



So the velocity of the ball increases from 0 to 40 because
There is a force applied to it during a time 0.005s

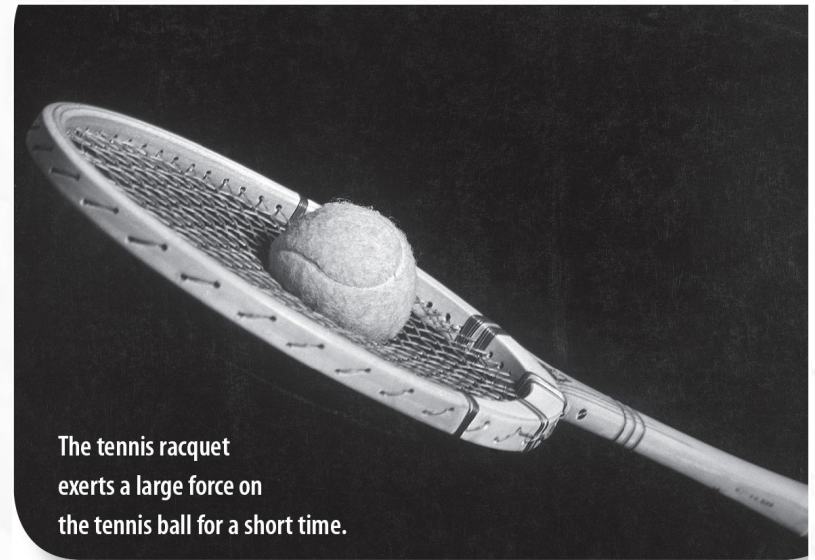
$$F\Delta t = m\Delta V$$

Let's estimate the average force on a tennis ball as it is served. The ball's mass is 0.06 kg and it leaves the racquet with a speed of 40 m/s. High-speed photography indicates that the contact time is about 5 milliseconds.

$$0.06(40-0) = F(0.005)$$

$$F = 480\text{N}$$

(about the weight of 48kg or 96pounds)

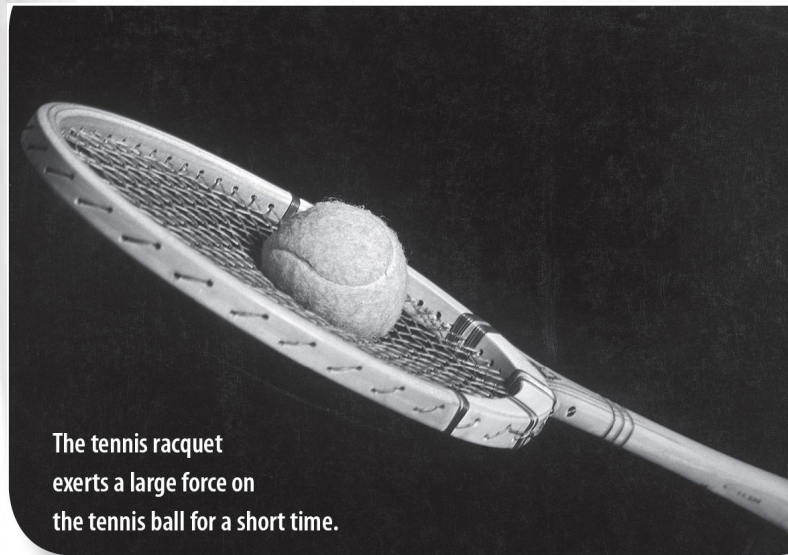


Why do you think you should have a good follow through?

You increase the time of contact for the same force F (same muscle). What happens to the speed of the ball ? (that is to its change of momentum ?)

Why a pitcher keep the baseball as long as possible when throwing ?

- This tells why we must exert a force to stop an object or get it to move.
- To stop a moving object, we have to bring its momentum to zero.
- To start moving an object, we have to impart some momentum to it.



In reality the force applied to the Ball by the tennis is not constant . It increases and decreases. The formula is in fact:

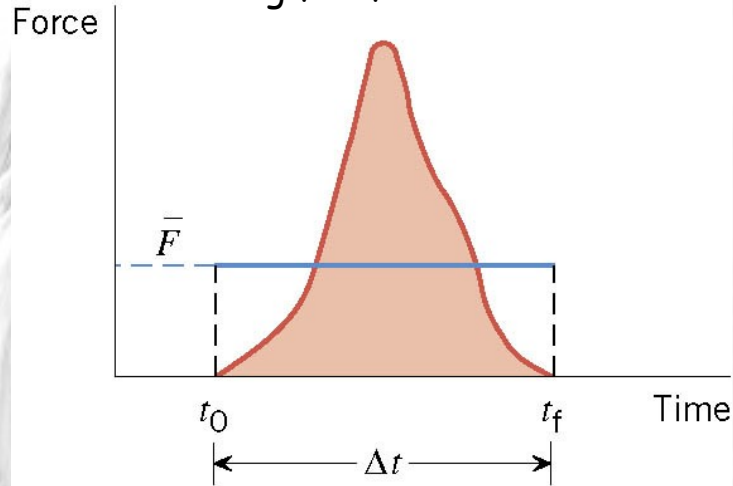
$$m\Delta v = \overline{F}\Delta t$$

Change in momentum = impulse

$\overline{F}$ Is the average force applied during the time Δt

$\overline{F}$ Is the average force applied during the time Δt

See gif in folder



$$m\Delta V = \overline{F}\Delta t$$

$$\Delta \vec{p} = \vec{J}$$

$$\Delta \vec{p} = m\vec{v}_f - m\vec{v}_i$$

Impulse = area under the curve
Force can change with time..

$$J = \int F dt = \overline{F}\Delta t$$

The minute of
calculus to prove →

$$\int_{t_1}^{t_2} \vec{F} dt = m \int_{t_1}^{t_2} \vec{a} dt$$

$$\int_{t_1}^{t_2} \vec{F} dt = m(\vec{v}_2 - \vec{v}_1)$$

$$\int_{t_1}^{t_2} \vec{F} dt = (m\vec{v}_2) - (m\vec{v}_1)$$

$$\overline{F}\Delta t = \Delta p = p' - p$$

If you integrate over time: impulse-momentum theorem
If you integrate over distance: kinetic-work theorem

$$\int_{x_1}^{x_2} F(x) dx = \frac{1}{2}m(v_2^2 - v_1^2)$$

IMPULSE-MOMENTUM THEOREM

And if more than 1 force act on an object, we need
To find first the net force before using the theorem
Called the impulse-momentum theorem;

When a net force acts on an object, the impulse of
this force is equal to the change in the momentum
of the object

impulse

$$\left(\sum \overline{F}\right)\Delta t = m(V_f - V_i) = mV_f - mV_i$$

final momentum

initial momentum

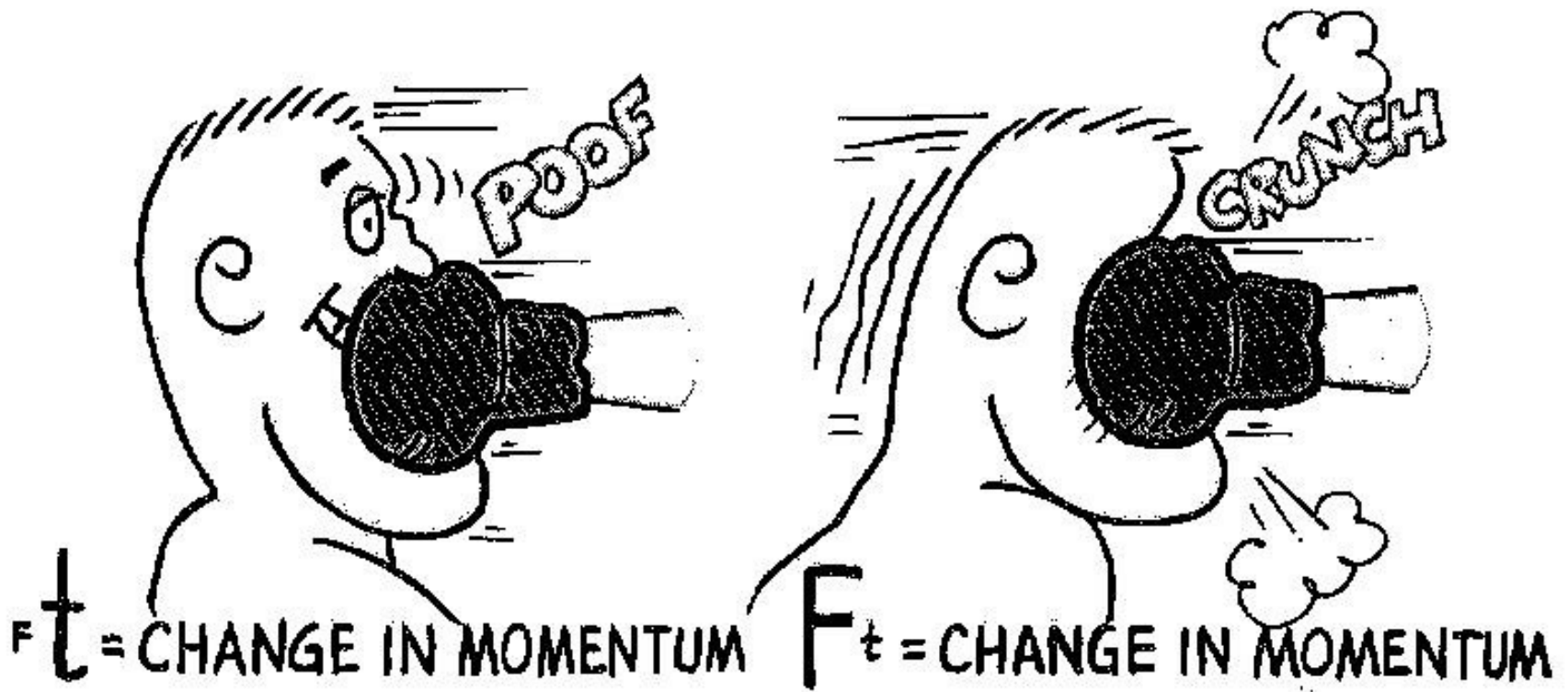
**BUT LETS GO BACK TO THE BIG IDEA
behind this formula:**

- It also tells us that we can change the motion using various forces and time intervals:

$$F\Delta t = m\Delta V$$

- Use a large force for a short time, or
- Use a small force for a long time.

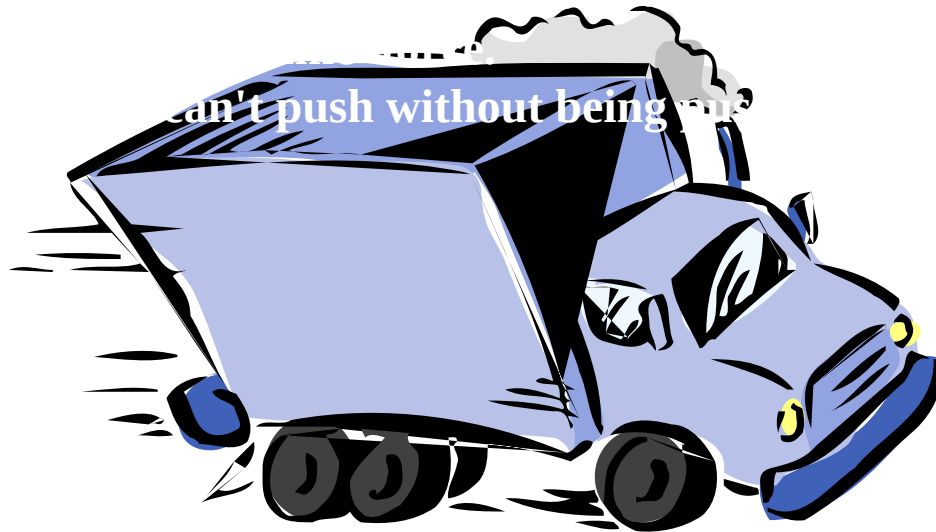
Change in motion = change in momentum



SO WHY DO YOU THINK a boxer that is getting punched needs to “roll over” ?

He moves away when the glove hits versus staying put ?

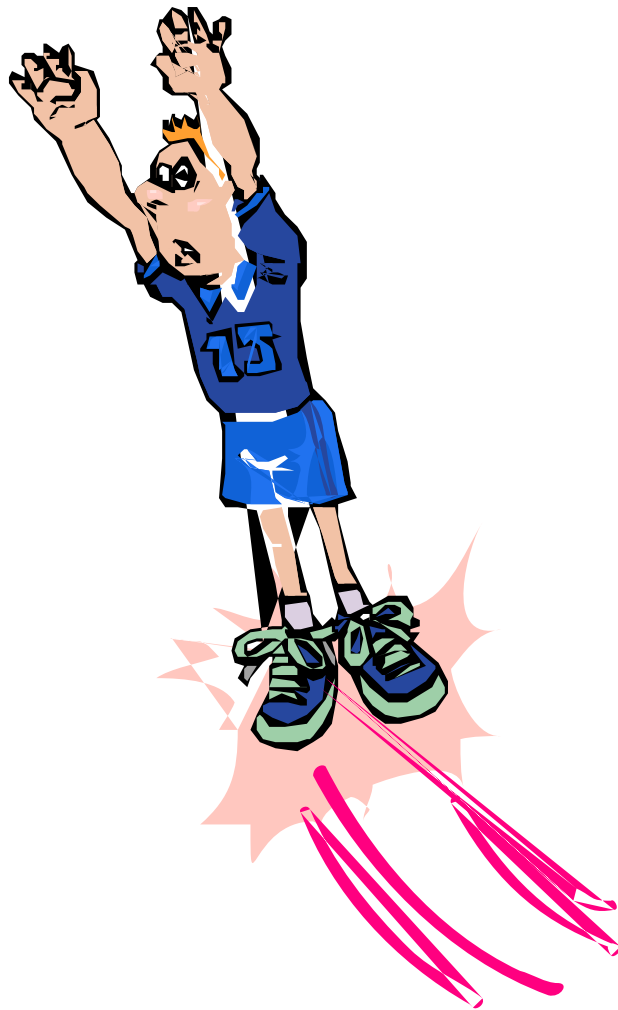
In both cases, the **change in momentum is the same**



A truck can be stopped by a haystack
Or by a brick of wall. **The change of motion**
Is the same. But what about time and force?
Which situation would you prefer ?

$$m\Delta v = \overline{F}\Delta t \quad \text{or} \quad m\Delta v = \overline{F}\Delta t$$





When you jump and you come
Back to the ground,
why don't you keep the
Legs stiff instead of bending them ?

Change in motion means a force is applied to the ground so
The ground applies the same force on the jumper



Calculating the Change of Momentum

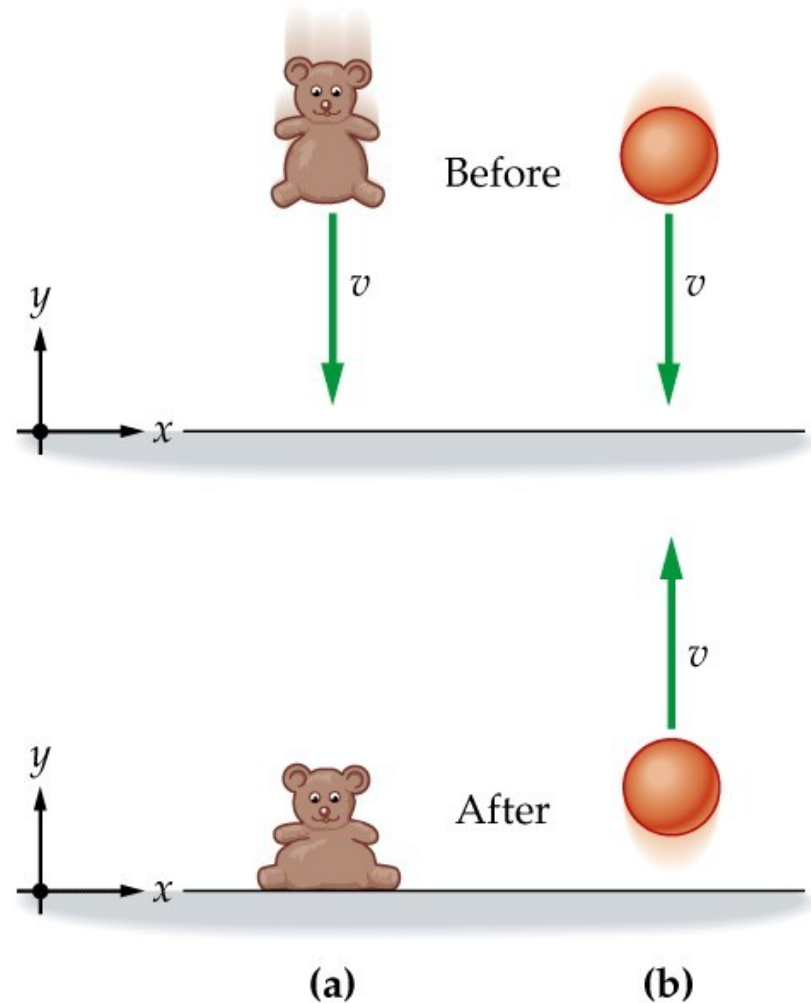
$$\begin{aligned}\Delta \vec{p} &= \vec{p}_{after} - \vec{p}_{before} \\ &= m\vec{v}_{after} - m\vec{v}_{before} \\ &= m(\vec{v}_{after} - \vec{v}_{before})\end{aligned}$$

For the teddy bear

$$\Delta p = m[0 - (-v)] = mv$$

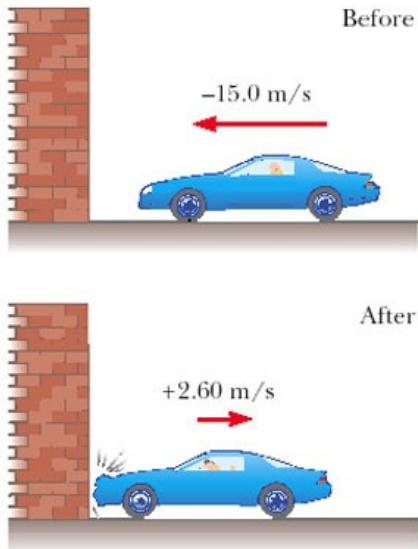
For the bouncing ball

$$\Delta p = m[v - (-v)] = 2mv$$



How Good Are the Bumpers?

- In a crash test, a car of mass 1.5×10^3 kg collides with a wall and rebounds as in figure. The initial and final velocities of the car are $v_i = -15$ m/s and $v_f = 2.6$ m/s, respectively. If the collision lasts for 0.15 s, find
- the impulse delivered to the car due to the collision
 - the size and direction of the average force exerted on the car



© 2006 Brooks/Cole - Thomson



How Good Are the Bumpers?

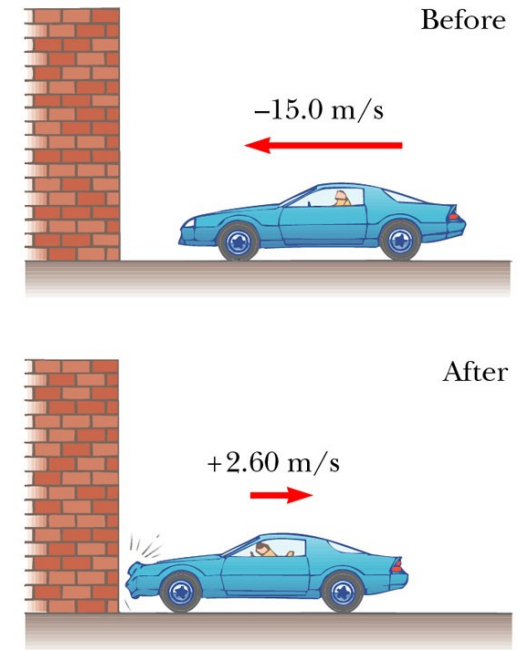
- In a crash test, a car of mass $1.5 \times 10^3 \text{ kg}$ collides with a wall and rebounds as in figure. The initial and final velocities of the car are $v_i = -15 \text{ m/s}$ and $v_f = 2.6 \text{ m/s}$, respectively. If the collision lasts for 0.15 s , find
- the impulse delivered to the car due to the collision
 - the size and direction of the average force exerted on the car

$$p_i = mv_i = (1.5 \times 10^3 \text{ kg})(-15 \text{ m/s}) = -2.25 \times 10^4 \text{ kg} \cdot \text{m/s}$$

$$p_f = mv_f = (1.5 \times 10^3 \text{ kg})(+2.6 \text{ m/s}) = +0.39 \times 10^4 \text{ kg} \cdot \text{m/s}$$

$$\begin{aligned} I &= p_f - p_i = mv_f - mv_i \\ &= (0.39 \times 10^4 \text{ kg} \cdot \text{m/s}) - (-2.25 \times 10^4 \text{ kg} \cdot \text{m/s}) \\ &= 2.64 \times 10^4 \text{ kg} \cdot \text{m/s} \end{aligned}$$

$$F_{av} = \frac{\Delta p}{\Delta t} = \frac{I}{\Delta t} = \frac{2.64 \times 10^4 \text{ kg} \cdot \text{m/s}}{0.15 \text{ s}} = 1.76 \times 10^5 \text{ N}$$



© 2006 Brooks/Cole - Thomson



Impulse-Momentum Theorem

- A child bounces a 100 g superball on the sidewalk. The velocity of the superball changes from 10 m/s downward to 10 m/s upward. If the contact time with the sidewalk is 0.1s, what is the magnitude of the impulse imparted to the superball?

- (A) 0
(B) 2 kg-m/s
(C) 20 kg-m/s
(D) 200 kg-m/s
(E) 2000 kg-m/s

$$\vec{I} = \Delta \vec{p} = m \vec{v}_f - m \vec{v}_i$$

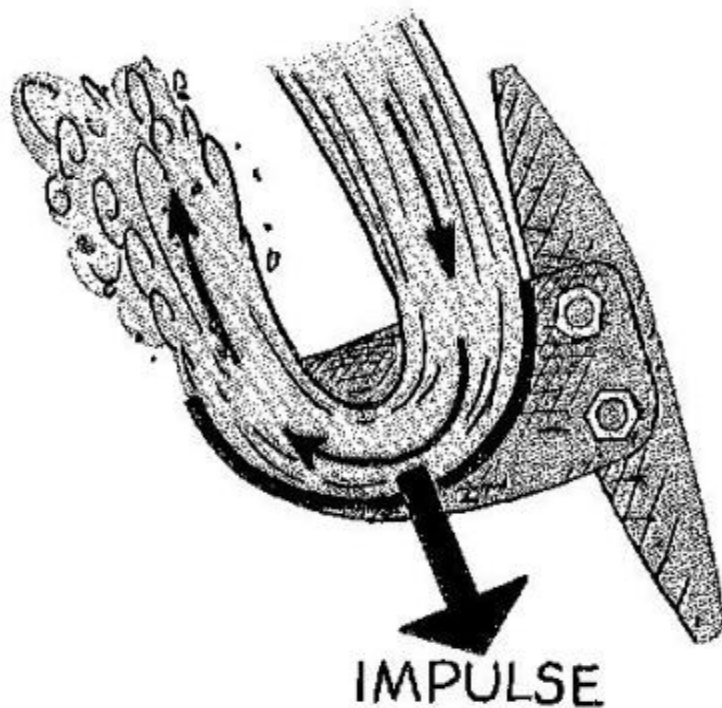


Impulse-Momentum Theorem 2

- A child bounces a 100 g superball on the sidewalk. The velocity of the superball changes from 10 m/s downward to 10 m/s upward. If the contact time with the sidewalk is 0.1s, what is the magnitude of the force between the sidewalk and the superball?
- (A) 0
(B) 2 N
(C) 20 N
(D) 200 N
(E) 2000 N

$$\vec{F} = \frac{\vec{I}}{\Delta t} = \frac{\Delta \vec{p}}{\Delta t} = \frac{m \vec{v}_f - m \vec{v}_i}{\Delta t}$$





So why do you think this Pelton Wheel was a breakthrough for the Gold miners ?

Before they were using flat paddles.
(bouncing =
more change in momentum = more impulse)

Source: Paul Hewitt, conceptual Physics.

Pelton wheel

The Pelton wheel is an impulse-type water turbine. It was invented by Lester Allan Pelton in the 1870s. The Pelton wheel extracts energy from the impulse of moving water, as opposed to water's dead weight like the traditional overshot water wheel. [Wikipedia](#)



Conservation of momentum is a consequence of Newton's third law

CONSERVATION OF MOMENTUM ?

When 2 objects “collide” (interact)→
Momentum lost by one object = momentum gained by the second object
(we neglect friction, air resistance).
This is because action = - reaction.



They both feel the same force
But the consequence is not
the same. Consequence
Depends on mass

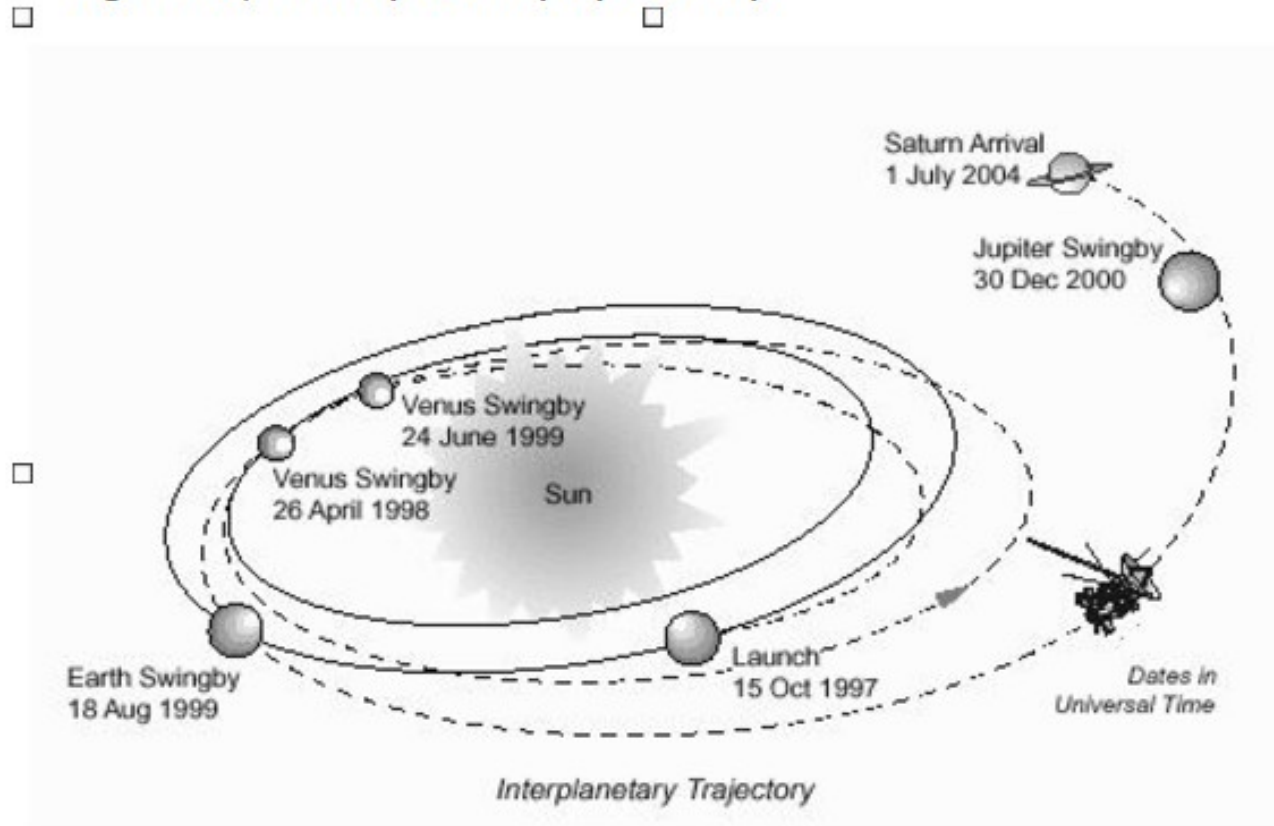
Divide by
time

$$M a = F = - m a$$
$$\downarrow$$
$$M \Delta v = - m \Delta v$$

Momentum lost by Baloo = momentum gained by Mowgli

DEMO Basket ball + tennis video

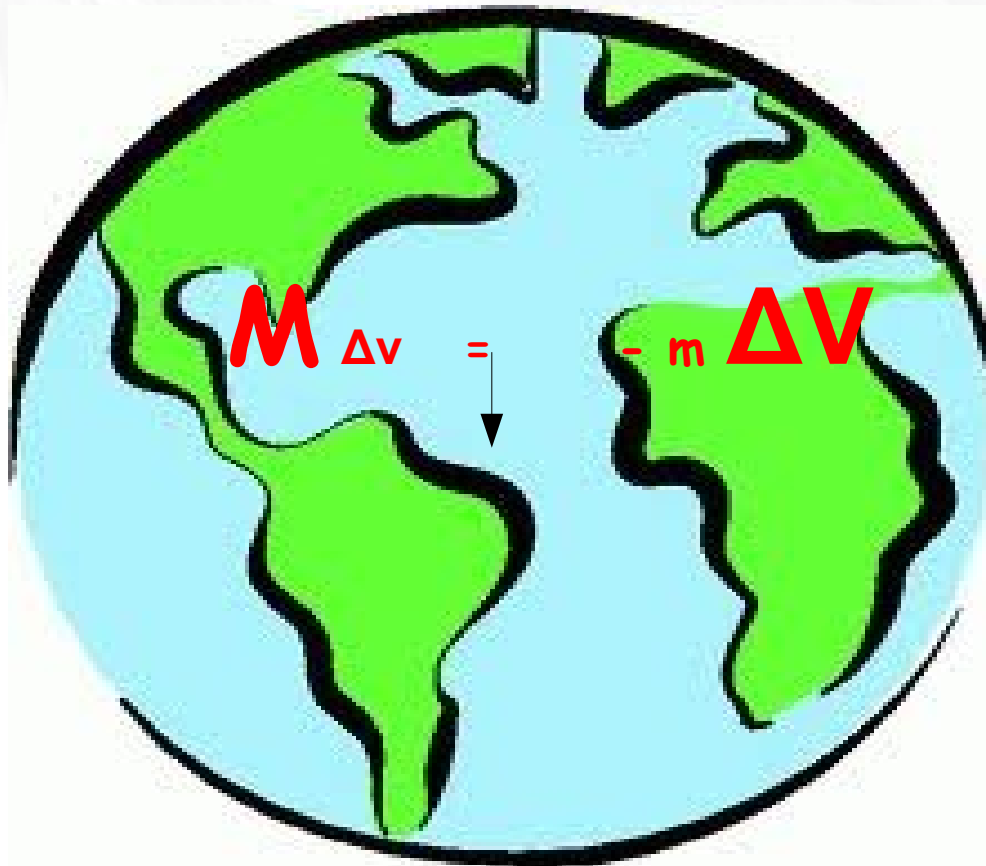
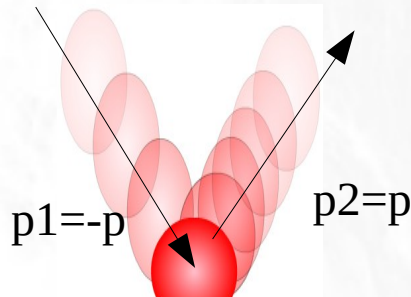
Slingshot effect= a spaceship gained momentum while
Passing by a planet. The planet loses momentum (but small acceleration because
Big mass). The spaceship speeds up.



The recoil of the Earth is tiny

$$\Delta p = 2p$$

Along the y-axis



When a ball bounces off the Earth,
The ball gained momentum.
The velocity changes direction.

So if $V_1 = -3$ (initial)

And $V_2 = 3$ (final)

That means the change in velocity
Is $+6$ or the gain in momentum is
 $6m$. $\Delta p = +6m = \text{gain}$.

That means that the Earth lost that
Amount too. But because the mass
Of the Earth is huge, the change in
Its velocity is tiny.

$$\Delta p = -M_{\text{huge}} \times \Delta v_{\text{small}} = -6m$$

All the change in momentum is in
The mass of the Earth

Note: the collision is elastic
So the kinetic energy is
Conserved. The kinetic energy
Is a scalar so the sign of the
Velocity does not matter:

$$KE_1 = 0.5 m V_1^2 = 0.5m(3)^2$$

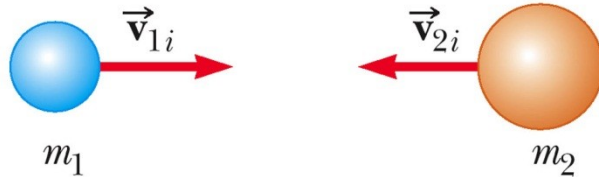
$$KE_2 = 0.5 m V_2^2 = 0.5 m (3)^2$$

$$\text{So } KE_2 = KE_1$$

$$\text{But } p_2 = -p_1$$

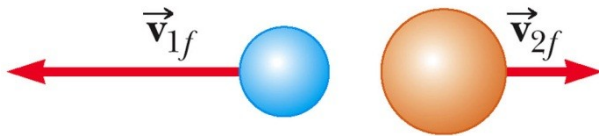
Conservation of Momentum

Before collision

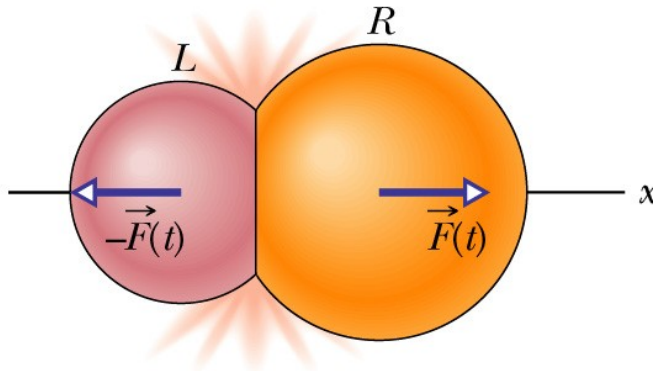


© 2008 Brooks/Cole - Thomson

After collision



© 2008 Brooks/Cole - Thomson



- Start from impulse-momentum theorem

$$\vec{F}_{21} \Delta t = m_1 \vec{v}_{1f} - m_1 \vec{v}_{1i}$$

$$\vec{F}_{12} \Delta t = m_2 \vec{v}_{2f} - m_2 \vec{v}_{2i}$$

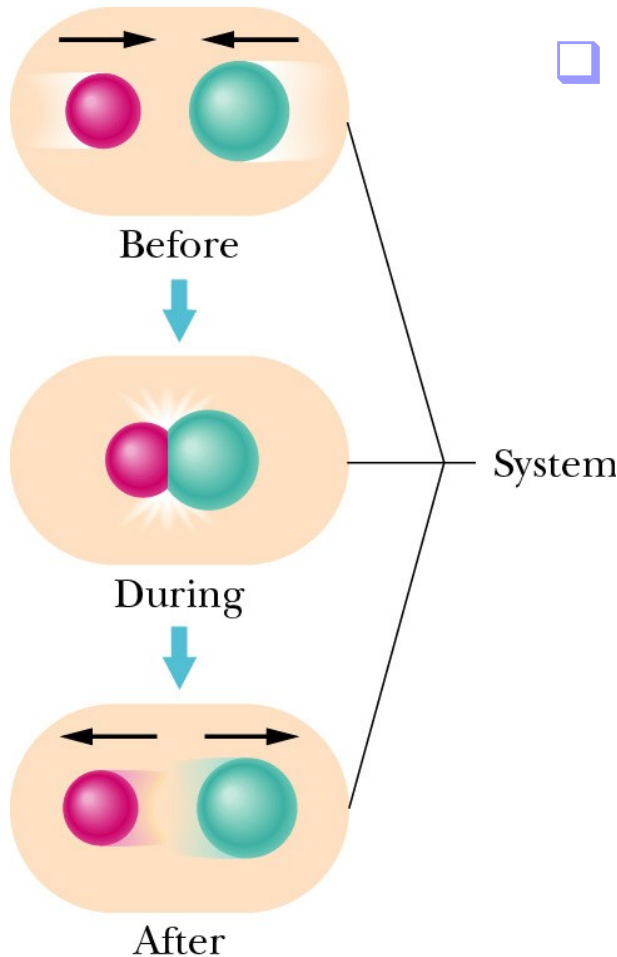
- Since $\vec{F}_{21} \Delta t = -\vec{F}_{12} \Delta t$

- Then $m_1 \vec{v}_{1f} - m_1 \vec{v}_{1i} = -(m_2 \vec{v}_{2f} - m_2 \vec{v}_{2i})$

- So $m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} = m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f}$



Conservation of Momentum



- In an isolated and closed system, the total momentum of the system remains constant in time.
 - Isolated system: no external forces
 - Closed system: no mass enters or leaves
 - The linear momentum of each colliding body may change
 - The total momentum P of the system cannot change.

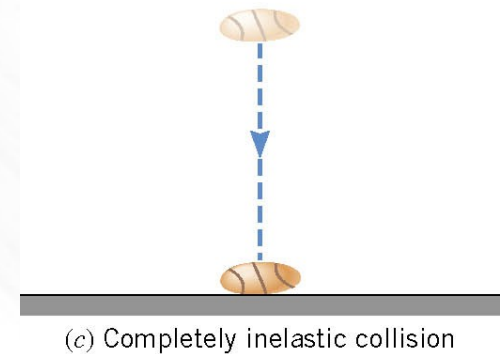
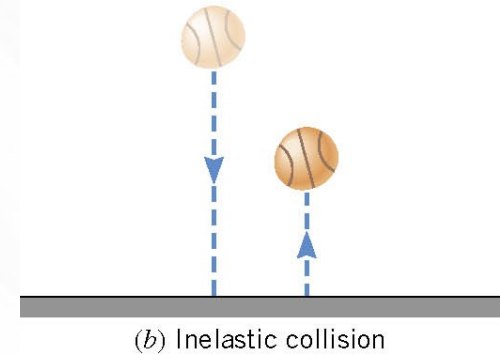
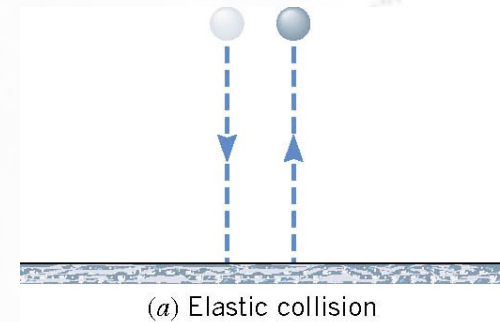


The total linear momentum is always conserved when two objects collide, provided they constitute an isolated system. That is if we neglect outside force (time of collision is short, we neglect friction).

But KE is not always conserved because KE can be turned to heat, noise, deformation.. From the work done by non conservative force. Momentum can only be changed in momentum.

Elastic collision -- One in which the total kinetic energy of the system after the collision is equal to the total kinetic energy before the collision.
!! in 1D with 2 objects. The relative velocity before = Relative velocity after.

Inelastic collision -- One in which the total kinetic energy of the system after the collision is not equal to the total kinetic energy before the collision; if the objects stick together after colliding, the collision is said to be completely inelastic.



Types of Collisions

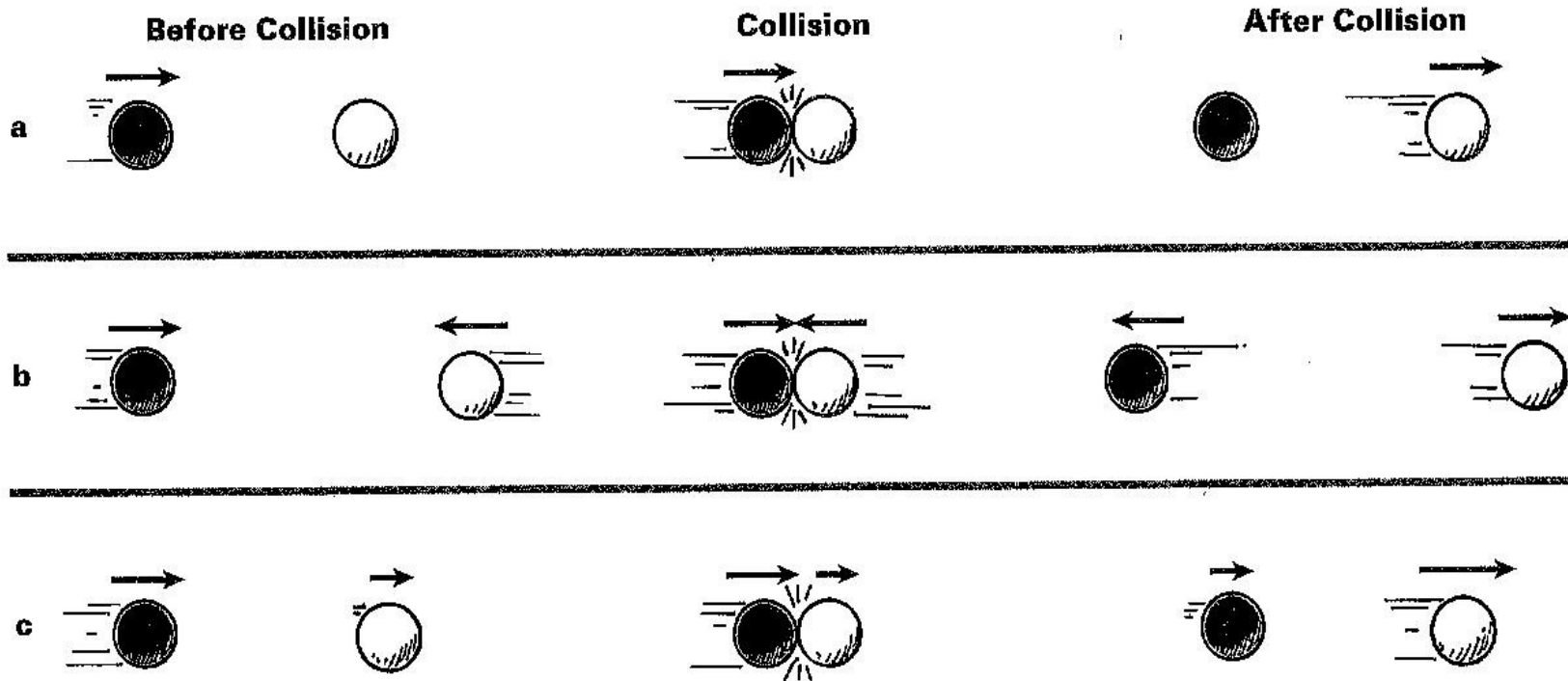
- ❑ Momentum is conserved in any collision
- ❑ **Inelastic collisions:** *rubber ball and hard ball*
 - Kinetic energy is not conserved
 - **Perfectly inelastic** collisions occur when the objects stick together
- ❑ **Elastic collisions:** *billiard ball*
 - both momentum and kinetic energy are conserved
- ❑ **Actual collisions**
 - Most collisions fall between elastic and perfectly inelastic collisions



Elastic collision = hit and bounce / hit and stop / hit and transfer momentum

We suppose the mass the same. **KE is conserved.**

The momentum lost by 1 object = momentum gained by the second object



Describe each collision. Is the momentum transferred from one ball to the other ?

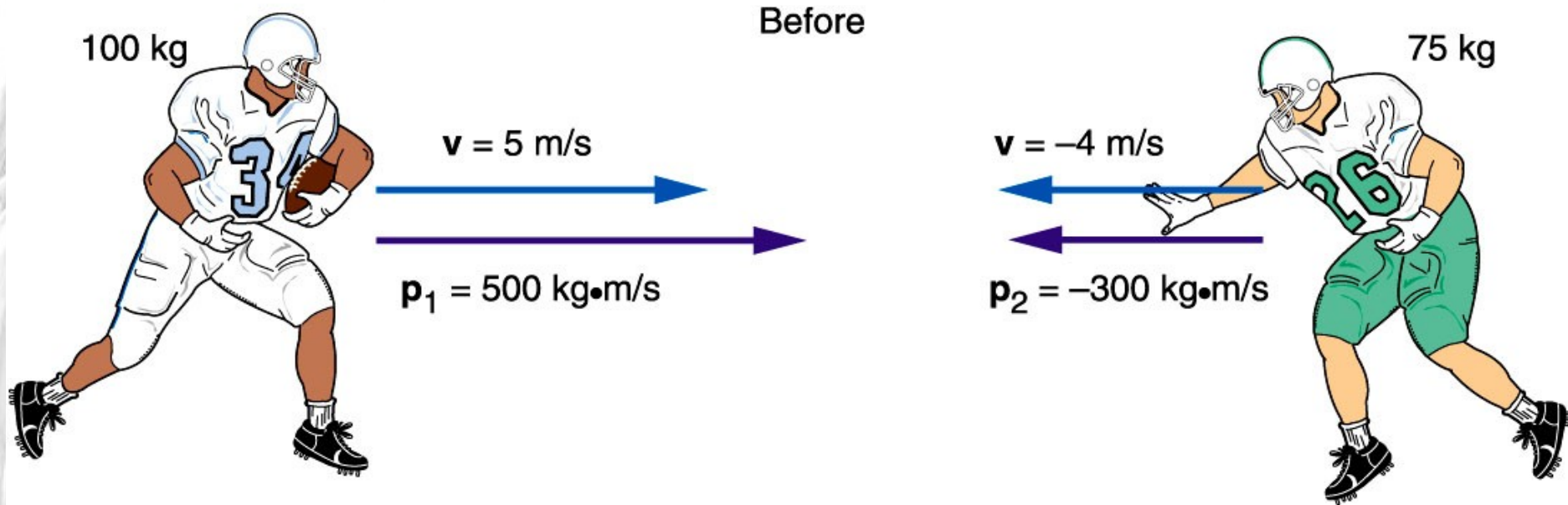
!!! the collisions are elastic $\rightarrow$ the relative velocity stays the same !!

1 $\rightarrow$ relative velocity is V

2 $\rightarrow$ relative velocity is $2V$

3 $\rightarrow$ relative velocity is $V_2 - V_1 = - (V_2' - V_1')$.

First let's investigate inelastic collision: hit and stick

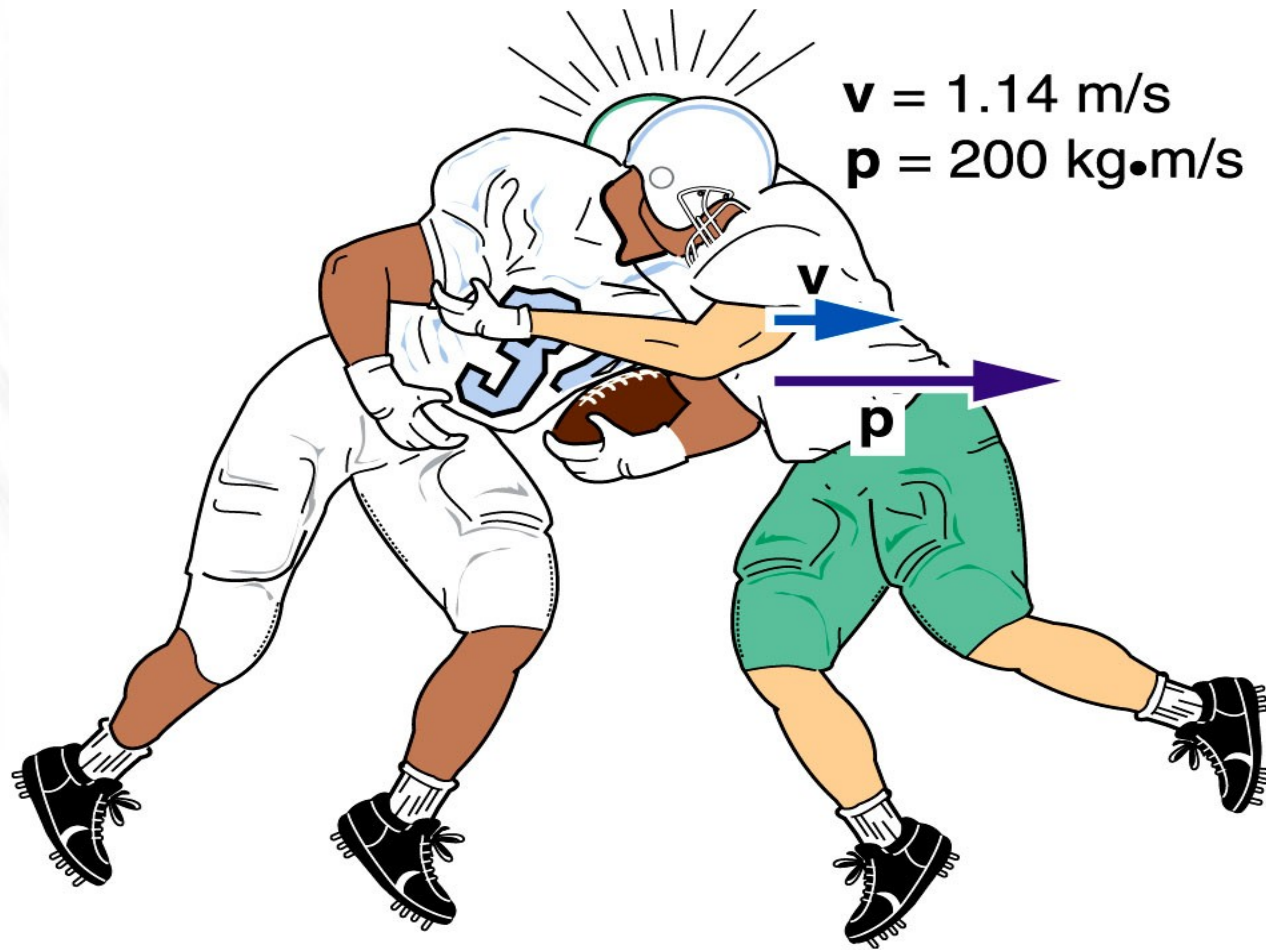


Video sport

THE 2 PLAYERS STICK TO EACH OTHER AFTER COLLISION

- 1) Find the FINAL VELOCITY (which way are they going)
- 2) the change of momentum for each player
- 3) the momentum before and after collision

After



Kinetic energy is not conserved.

Energy went to broken bones, bruises and heat and sound

What do you think happens if they both have the same momentum before collision?

See simulations https://www.walter-fendt.de/html5/phen/collision_en.htm

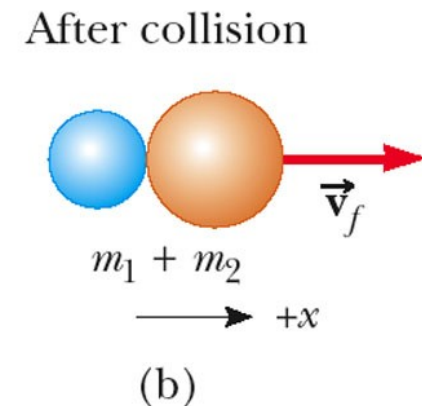
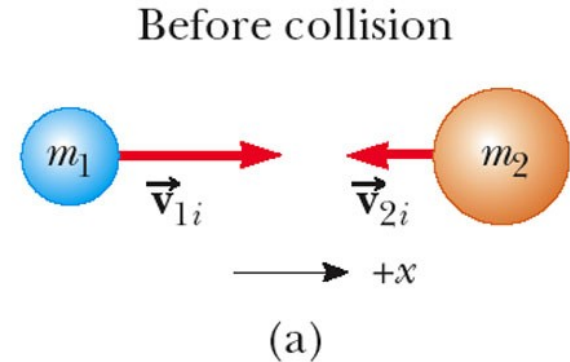
More about Perfectly Inelastic Collisions

- When two objects stick together after the collision, they have undergone a perfectly inelastic collision
- Conservation of momentum

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$$

- Kinetic energy is **NOT** conserved

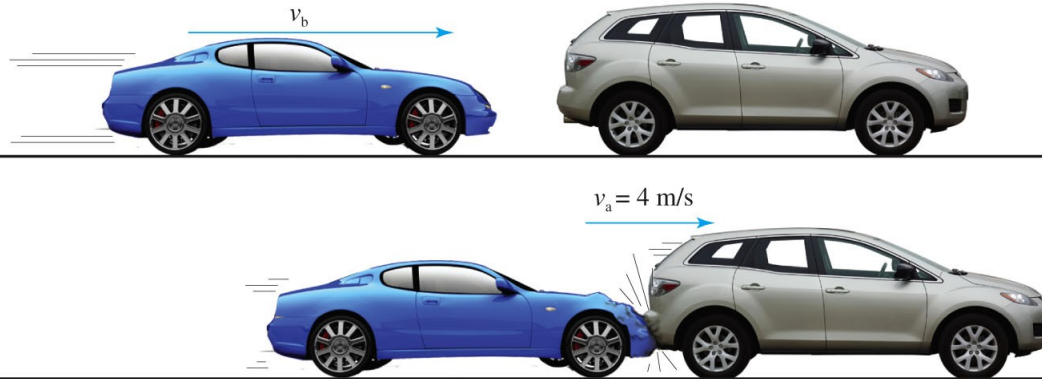


© 2006 Brooks/Cole - Thomson



Example #2

A 1,000 kg car (car 1) runs into the rear of a stopped car (car 2) that has a mass of 1,500 kg. Immediately after the collision, the cars are hooked together and have a speed of 4 m/s. What was the speed of car 1 just before the collision?



Use: total momentum before = total momentum after

Or $1000V_1 + 1500(0) = 1000(4) + 1500(4)$

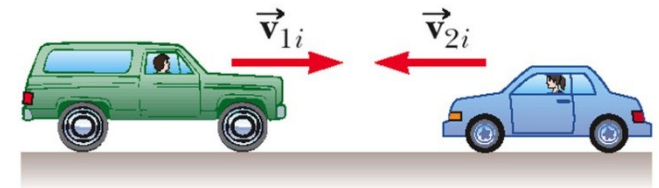
$$1000 V_1 = 2500(4)$$

$$V_1 = 2.5(4) = 10 \text{ m/s}$$

**Note that in a “hit and stick” situation, kinetic energy is not conserved. Compute the kinetic energy of the System before (1000kg@10m/s) and after (2500kg@4m/s)
How much energy was lost ? What is the work done by the friction ?**

An SUV Versus a Compact

- An SUV with mass 1.80×10^3 kg is travelling eastbound at $+15.0$ m/s, while a compact car with mass 9.00×10^2 kg is travelling westbound at -15.0 m/s. The cars collide head-on, becoming entangled.



(a)



(b)

© 2006 Brooks/Cole - Thomson



An SUV Versus a Compact

(a) Find the speed of the entangled cars after the collision.

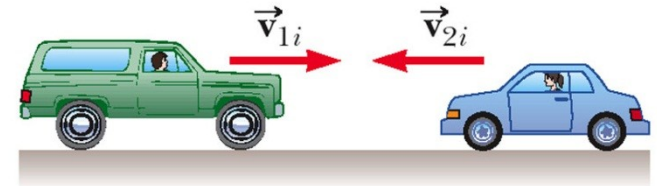
$$m_1 = 1.80 \times 10^3 \text{ kg}, v_{1i} = +15 \text{ m/s}$$
$$m_2 = 9.00 \times 10^2 \text{ kg}, v_{2i} = -15 \text{ m/s}$$

$$p_i = p_f$$

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$$

$$v_f = +5.00 \text{ m/s}$$



(a)



(b)

© 2006 Brooks/Cole - Thomson



An SUV Versus a Compact

(b) Find the change in the velocity of each car.

$$v_f = +5.00 \text{ m/s}$$

$$\Delta v_1 = v_f - v_{1i} = -10.0 \text{ m/s}$$

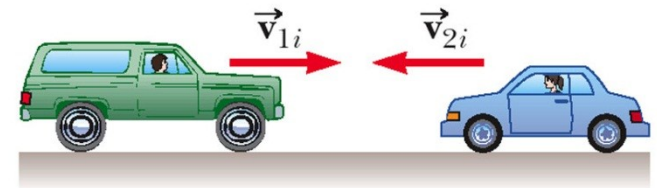
$$\Delta v_2 = v_f - v_{2i} = +20.0 \text{ m/s}$$

$$m_1 \Delta v_1 = m_1 (v_f - v_{1i}) = -1.8 \times 10^4 \text{ kg} \cdot \text{m/s}$$

$$m_2 \Delta v_2 = m_2 (v_f - v_{2i}) = +1.8 \times 10^4 \text{ kg} \cdot \text{m/s}$$

$$m_1 \Delta v_1 + m_2 \Delta v_2 = 0$$

$$m_1 = 1.80 \times 10^3 \text{ kg}, v_{1i} = +15 \text{ m/s}$$
$$m_2 = 9.00 \times 10^2 \text{ kg}, v_{2i} = -15 \text{ m/s}$$



(a)



(b)

© 2006 Brooks/Cole - Thomson



An SUV Versus a Compact

- (c) Find the change in the kinetic energy of the system consisting of both cars.

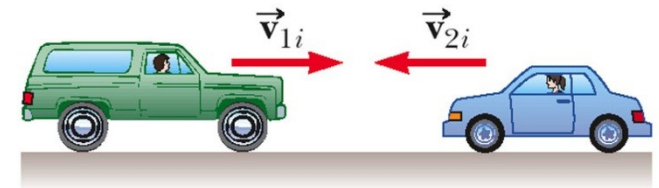
$$m_1 = 1.80 \times 10^3 \text{ kg}, v_{1i} = +15 \text{ m/s}$$
$$m_2 = 9.00 \times 10^2 \text{ kg}, v_{2i} = -15 \text{ m/s}$$

$$v_f = +5.00 \text{ m/s}$$

$$KE_i = \frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = 3.04 \times 10^5 \text{ J}$$

$$KE_f = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 = 3.38 \times 10^4 \text{ J}$$

$$\Delta KE = KE_f - KE_i = -2.70 \times 10^5 \text{ J}$$



(a)



(b)

© 2006 Brooks/Cole - Thomson



Clicker Question 8



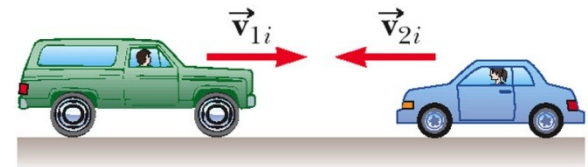
- A car of mass m travels with a speed v and collides with a parked truck, whose mass is $3m$. If the bumpers lock together in this head-on inelastic collision, what fraction of the initial kinetic energy is lost in this collision process?
- A. $1/4$
B. $1/3$
C. $1/2$
D. $3/4$
E. $2/3$



Conservation of Momentum

- An object of mass m moves to the right with a speed v . It collides head-on with an object of mass $3m$ moving with speed $v/3$ in the opposite direction. If the two objects stick together, what is the speed of the combined object, of mass $4m$, after the collision?

- (A) 0
- (B) $v/2$
- (C) v
- (D) $2v$
- (E) $4v$



(a)



(b)

Notebook interesting

© 2006 Brooks/Cole - Thomson



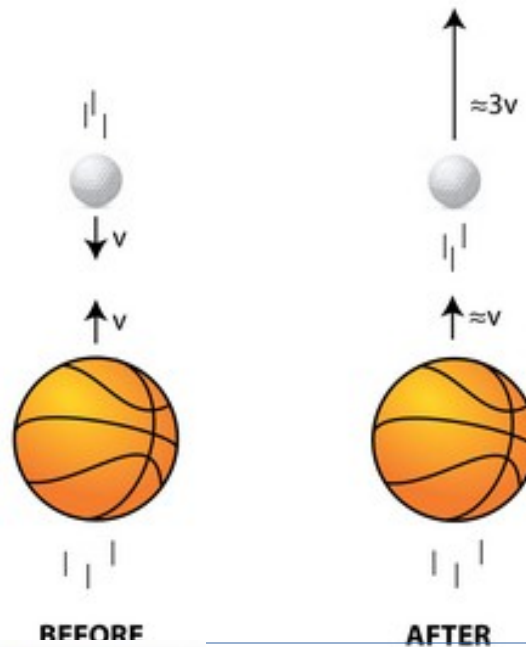


$$PE = mgh$$



Demo Will
And trampoline,

First let's investigate elastic collision: hit and stick
First play with simulations.

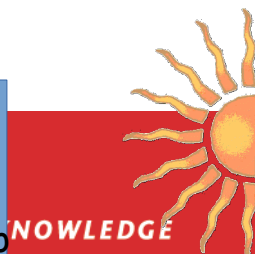


The collision is elastic
→ relative velocity is conserved
And the mass $M \gg m$
→ before collision the relative velocity is $2V$
→ after the collision
The relative velocity is $2V$

So velocity of ping pong is $3V \rightarrow 9$ times the initial KE.
→ 9 times the PE (KE $\rightarrow$ PE)
So 9 times the height !!

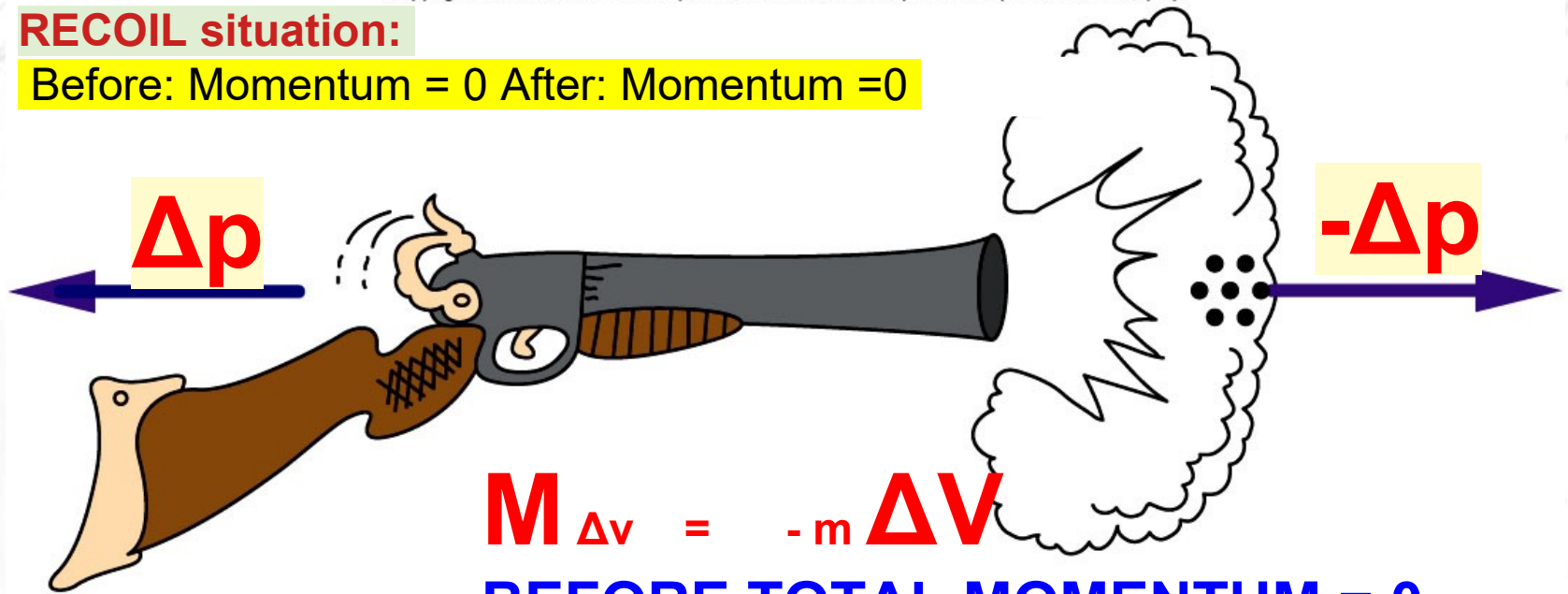


Crab nebula



RECOIL situation:

Before: Momentum = 0 After: Momentum = 0



$$M \Delta v = -m \Delta V$$

BEFORE TOTAL MOMENTUM = 0

AFTER TOTAL MOMENTUM = 0

$$0 = p_1' + p_2'$$

$$P_1' = -p_2' \quad m_1 V_1' = -m_2 V_2'$$

$$m_2/m_1 = -V_1'/V_2'$$

THE FORCE applied to the bullets and to the rifle is the same.

BUT which one undergo the larger acceleration ?

IS there a kick BACK ?

WHAT do YOU NEED to do so your shoulder DOES NOT BREAK ?



Recoil situation

MOMENTUM before = _____

The mass of granny = 100kg

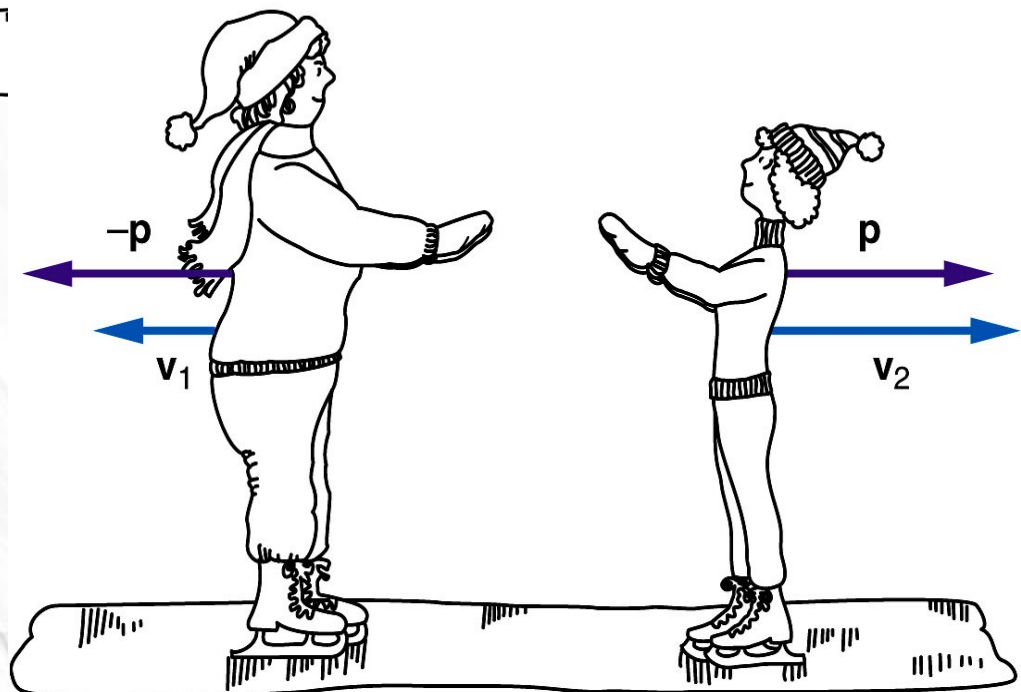
The mass of the girl = 50kg

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

Recoil

100 kg

50 kg



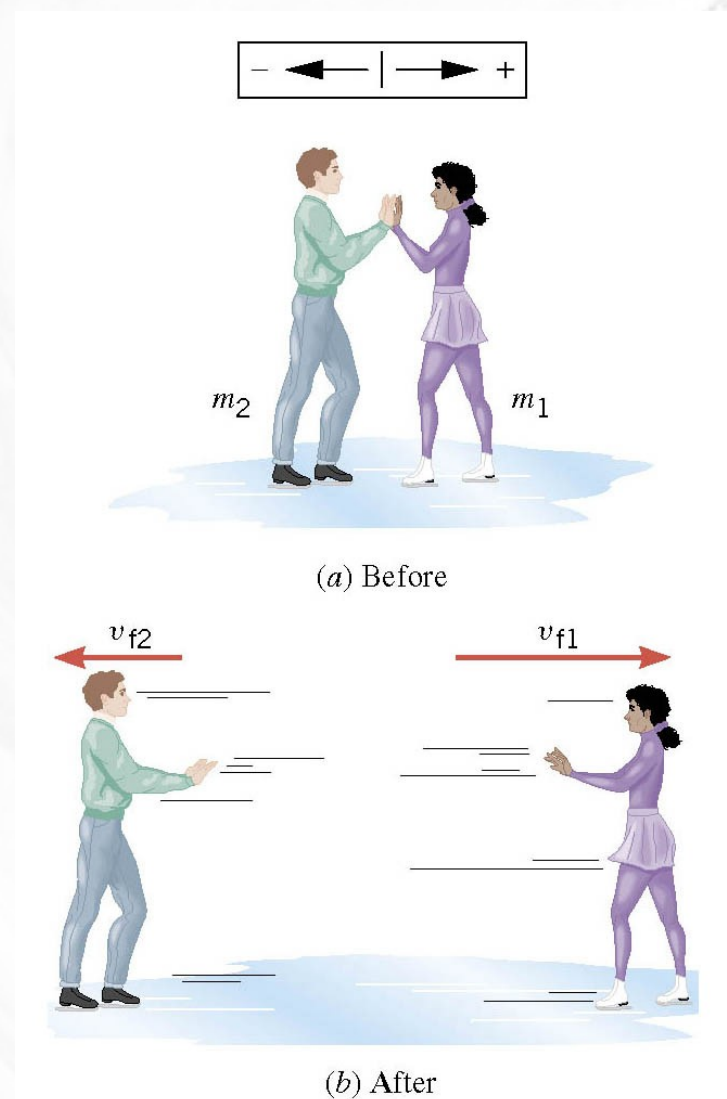
MOMENTUM after = _____

Compare the velocity of
Granny versus velocity of the girl

Example 6 Ice Skaters

Starting from rest, two skaters push off against each other on ice where friction is negligible.

One is a 54-kg woman and one is a 88-kg man. The woman moves away with a speed of +2.5 m/s. Find the recoil velocity of the man.

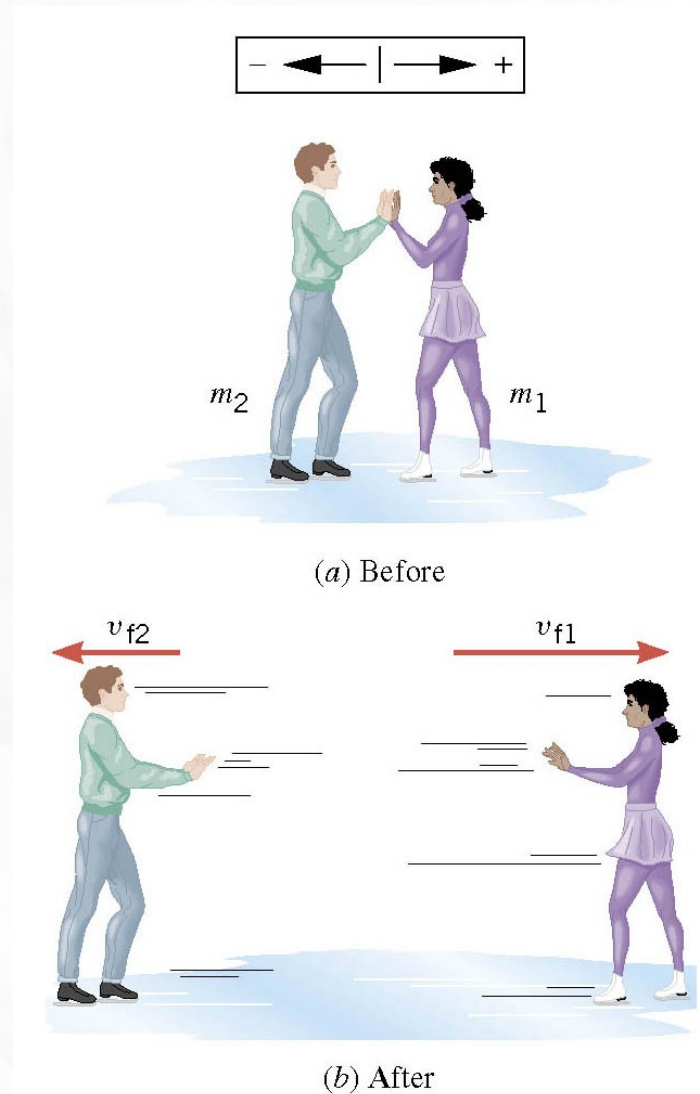


$$\vec{\mathbf{P}}_f = \vec{\mathbf{P}}_o$$

$$m_1 v_{f1} + m_2 v_{f2} = 0$$

$$v_{f2} = -\frac{m_1 v_{f1}}{m_2}$$

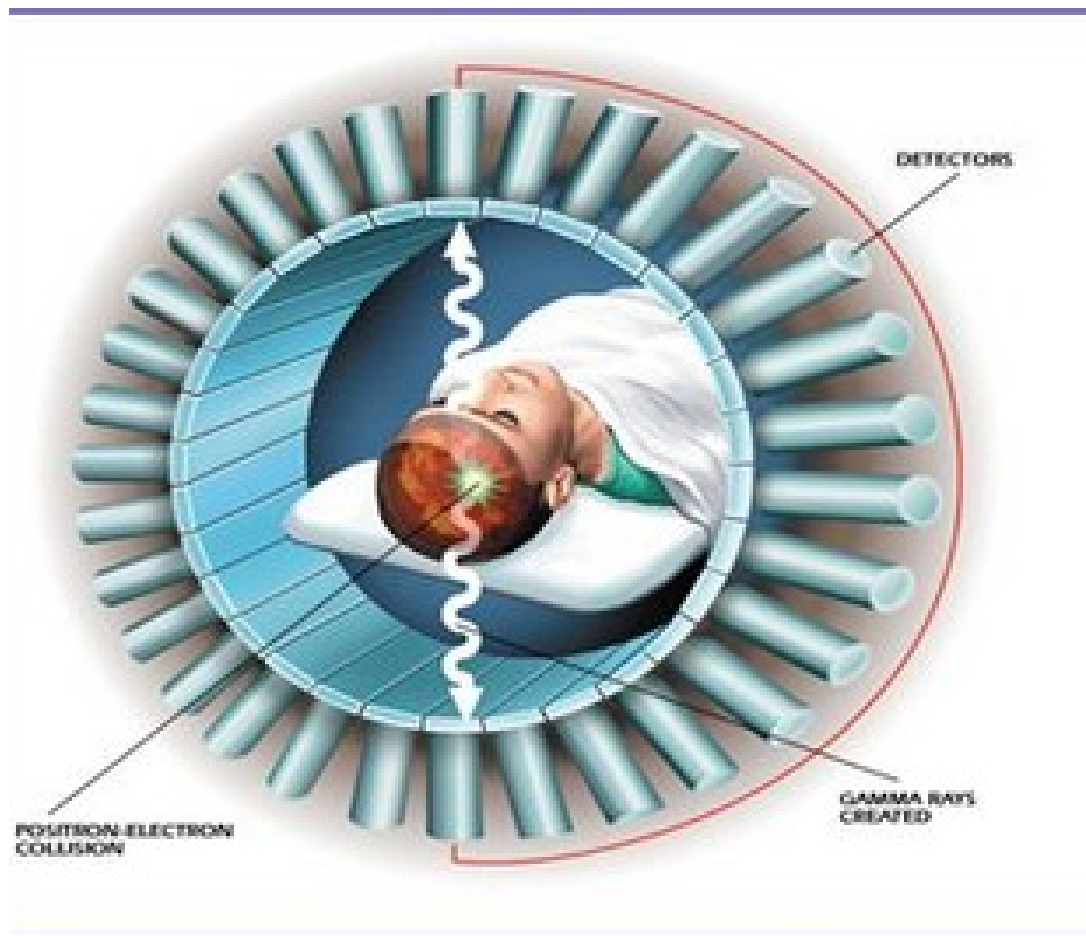
$$v_{f2} = -\frac{(54 \text{ kg})(+2.5 \text{ m/s})}{88 \text{ kg}} = -1.5 \text{ m/s}$$



Conservation of Momentum

- A 100 kg man and 50 kg woman on ice skates stand facing each other. If the woman pushes the man backwards so that his final speed is 1 m/s, at what speed does she recoil?
(A) 0
(B) 0.5 m/s
(C) 1 m/s
(D) 1.414 m/s
(E) 2 m/s





The Archer

□ An archer stands at rest on frictionless ice and fires a 0.5-kg arrow horizontally at 50.0 m/s. The combined mass of the archer and bow is 60.0 kg. With what velocity does the archer move across the ice after firing the arrow?

$$p_i = p_f$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$m_1 = 60.0 \text{ kg}, m_2 = 0.5 \text{ kg}, v_{1i} = v_{2i} = 0, v_{2f} = 50 \text{ m/s}, v_{1f} = ?$$

$$0 = m_1 v_{1f} + m_2 v_{2f}$$

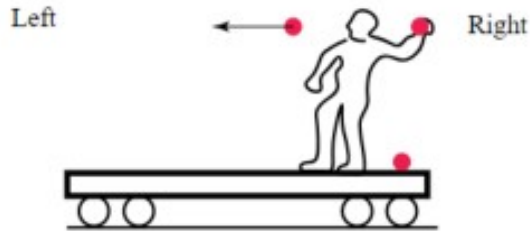
$$v_{1f} = -\frac{m_2}{m_1} v_{2f} = -\frac{0.5 \text{ kg}}{60.0 \text{ kg}} (50.0 \text{ m/s}) = -0.417 \text{ m/s}$$



© 2006 Brooks/Cole - Thomson



Study case: the rocket - recoil situation but $m(t)$



Conservation of momentum. What happens to the cart?

Imagine the person has a stack of tennis ball.

Throwing one by one faster what happens?

And if the rate at which they are thrown is faster

Suppose you are throwing the balls at a speed U (relative to cart) .

The total mass of the system (you + stack+ cart) is $m(t)$ and changes over time (as balls are ejected)

The mass of a ball is Δm (it was part of the system). The velocity of the cart increases by ΔV

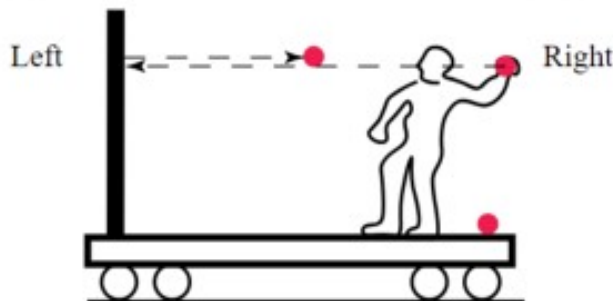
1) Momentum gain by a ball = - momentum lost by the system (cart+you+stack)

Find the thrust of the cart $m a$ (divide by Δt both sides)

2) Momentum gain by a ball = - momentum lost by the system (cart+you+stack)

Integrate.

Find the rocket equation. The final velocity V_f of the rocket as a function of V_i , U , m_o and m_f .



What now?



Same principle with a rocket. The fuel = tennis balls rocket propulsion

The rocket loses mass from burning fuel.

The thrust comes from ejecting a mass Δm of fuel
During a time Δt . U is the speed of exhaust gas
relative to rocket.

Momentum gained of rocket = momentum lost of gas
(see notebook)

$$M_{\Delta V}$$

$$M_{\Delta V} = - \Delta M U$$

$$dM/M = - dV/U \rightarrow \ln(M_f/M_i) = - 1/U (V_f - V_i)$$
$$V_f - V_i = U \ln(M_i/M_f)$$

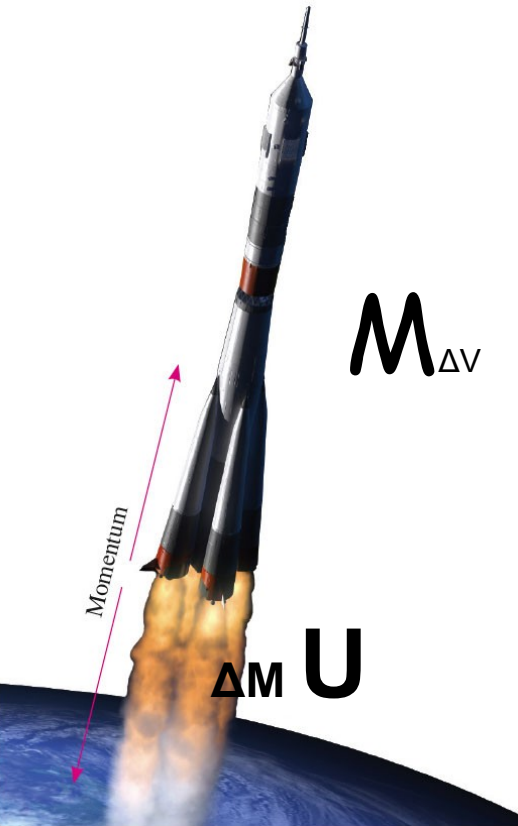
$$\text{THRUST} = M(t) \Delta V / \Delta t = - U \Delta M / \Delta t > 0$$

The thrust depends how fast the fuel is ejected and
How much fuel is ejected per unit time.

The true equation is:

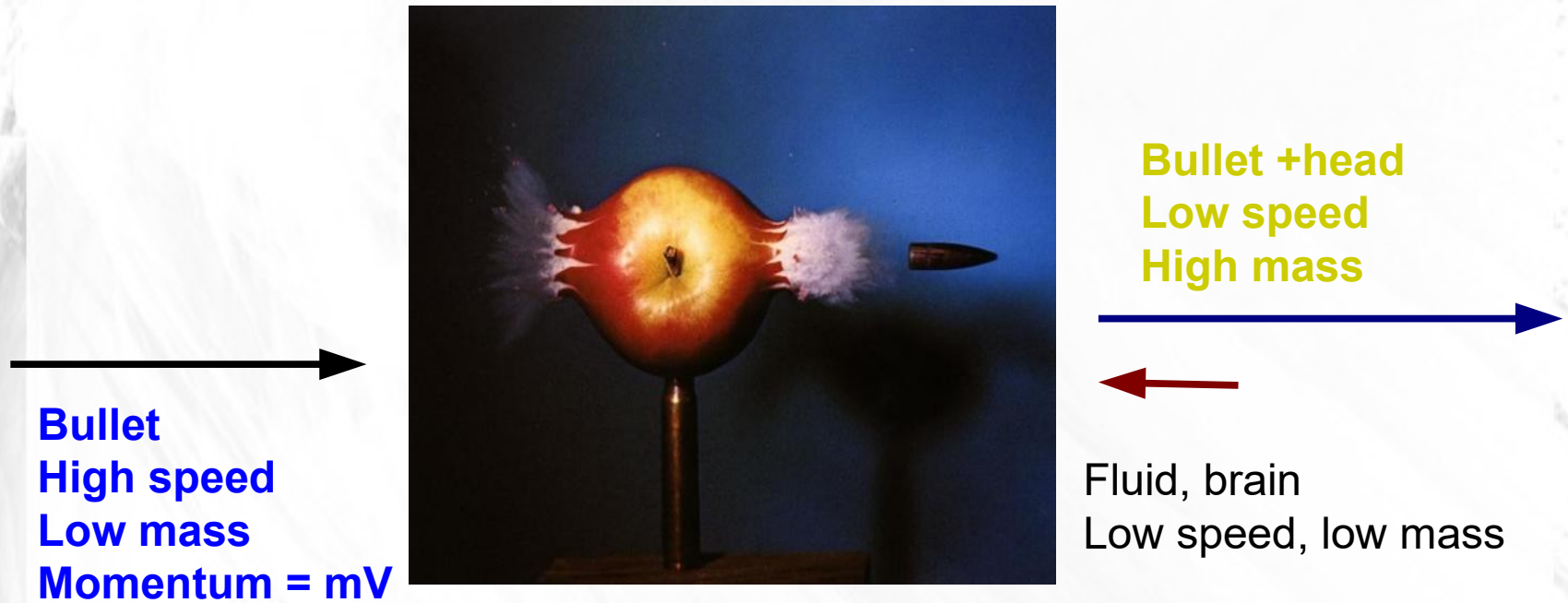
$$\text{Thrust} - mg = \text{mass_rocket}(t) \times \text{acceleration}$$

$$!! M(t) = M_i - (\Delta M / \Delta t) \times \text{time elapsed. (the mass changes)}$$



Textbook 8.15
8.16

Conservation of momentum





Luiz Alvarez: JFK and momentum

[Veronique Lankar](#)

All Sections

Nov 9 at 5:17pm

In the early 1960s, Nobel Prize-winning physicist Luis Alvarez conducted an experiment using a watermelon to investigate the Warren Commission's conclusion that three shots were fired during the assassination of President John F. Kennedy.

Alvarez believed that the Warren Commission's reconstruction of the events was flawed, and that a single shooter could have fired all three shots. To test his hypothesis, he set up an experiment in which he shot a watermelon with a high-powered rifle.

Alvarez took high-speed photographs of the watermelon as it was being shot. The photographs showed that the watermelon exploded upon impact, with material being expelled in both the forward and backward directions. This was consistent with Alvarez's understanding of the conservation of momentum, which states that the total momentum of a system must remain constant unless an external force acts on it.

Alvarez argued that the Warren Commission's reconstruction of the assassination was inconsistent with the conservation of momentum. If three shots had been fired, as the Warren Commission claimed, then the momentum of the bullets would have caused the president's body to be thrown backwards. However, the Zapruder film, which showed the assassination in slow motion, showed that the president's body was thrown forwards.

Alvarez's experiment was controversial, and his conclusions were challenged by other scientists. However, his work helped to raise doubts about the Warren Commission's findings and led to renewed calls for an investigation into the assassination.

1. **in class** . A fullback with a mass of 100 kg and a velocity of 3.5 m/s due west collides head-on with a Defensive back with a mass of 80kg and a velocity of 6m/s due east.

- A)What is the initial momentum of each player ?
- B)What is the total momentum of the system before the collision?
- C)If the stick together and external forces can be ignored , what direction will they be traveling Immediately after they collide ? What is the velocity?

2. **in class** . A Ice skater with a mass of 80kg pushed off against a second later with a mass of 32kg. Both Skaters are initially at rest.

- A) What is the total momentum of the system after they push off ?
- B) If the larger skater moves off with a speed of 3m/s, what is the corresponding speed of the smaller Skater ?

3. (HW) A rifle with a mass of 2.2kg fires a bullet with a mass of 5g(0.005kg) . The bullet moves with a Velocity of 600m/s after the rifle is fired !!! (1340 mph!!)

- A)What is the momentum of the bullet after the rifle is fired ? (hint: recoil situation !)
- B)If external forces acting on the riffle can be ignored, what is the recoil velocity of the rifle ? (use: equation of recoil).

4.(HW)A rocket ship at rest in space gives a short blast of its engine, firing 50kg of exhaust gas out the back ends (the gas) with an average velocity of 400m/s What is the change of momentum of the rocket during the Blast?

Hint: change of momentum of gas = change of momentum of rocket.

The gas has an initial speed of 0. (it is inside the rocket)

5. in class. A rail road car with a mass of 12,000kg collides and couples with a second car of mass 18,000kg That is initially at rest.

The first car is moving with a speed of 12m/s prior to the collision.

A)What is the initial momentum of the first car ?

B)If external forces can be ignored, what is the final velocity of the 2 railroad cars after they couple ?

6. **in class**. A fast ball thrown with a velocity of 40m/s (90mph) is struck by a baseball bat, and a line drive Comes back toward the pitcher with a velocity of 60m/s . The collision is between the bat and the ball. The ball is in contact with the bat for a time of just 0.04s . The baseball has a mass of 120g (0.12kg)

- A) What is the change in momentum of the baseball during this process ?
- B) What is the magnitude of the impulse required to produce this change in momentum ?
- C) What is the magnitude of the average force that acts on the baseball to produce this impulse ?
(impulse = change in momentum)

7. HW. A bullet is fired into a block of wood sitting on a block of ice. The bullet has an initial velocity of 500m/s and a mass of 0.005kg . The wooden block has a mass of 1.2kg and is initially at rest. The bullet remains embedded in the wood afterward. (so hit and stick situation)

- A) Assuming that momentum is conserved, find the velocity of the block of wood and bullet after the Collision. (so momentum before = momentum. $V_1' = V_2' = V'$)
- B) What is the magnitude of the impulse that acts on the block of wood in this process ?
- C) Does the change in momentum of the bullet equals that of the wood ?

8. **in class**. Consider 2 cases in which the same ball is thrown against a wall with the same initial velocity. In case A, the ball sticks to the wall and does not bounce. In case B, the ball bounces back with the Same speed that it came in with.

- A) In which of these examples is the change in momentum the largest ?
- B) Assuming that the time during which the momentum change takes place is approximately The same for these 2 cases, in which case is the larger average force involved ?
- C) Is the momentum conserved in this collision?

9. (review. HW) A car traveling at a speed of 18m/s (40mpg) crashes into a solid concrete Wall. The driver has a mass of 90kg .

A) What is the change in momentum of the driver as he comes to a stop ? ($V'=0$)

B) What impulse is required in order to produce this change in momentum ?

10. (HW) A 1500kg car traveling due North with a speed of 25m/s collides head-on with a 4500kg Truck traveling due South with a speed of 15m/s .

The 2 vehicles stick together after the collision

A) What is the total momentum of the system prior to the collision?

B) What is the velocity of the 2 vehicles after the collision ?

11. (HW) A 4000kg truck traveling with a velocity of 10m/s due north collide head-on with a 1200kg car Traveling with a velocity of 20m/s due South.

The 2 vehicles stick together after collision.

A) draw the situation (hit and stick)

B) What is the momentum of each vehicle prior to the collision ?

C) What are the size and direction of the total momentum of the 2 vehicles after they collide ? (so p_1+p_2)

12. (skip) For the 2 vehicles in 11. Sketch to scale the momentum vectors of the 2 vehicles

A) Prior to the collision.

B) Add the 2 vectors on your sketch graphically.

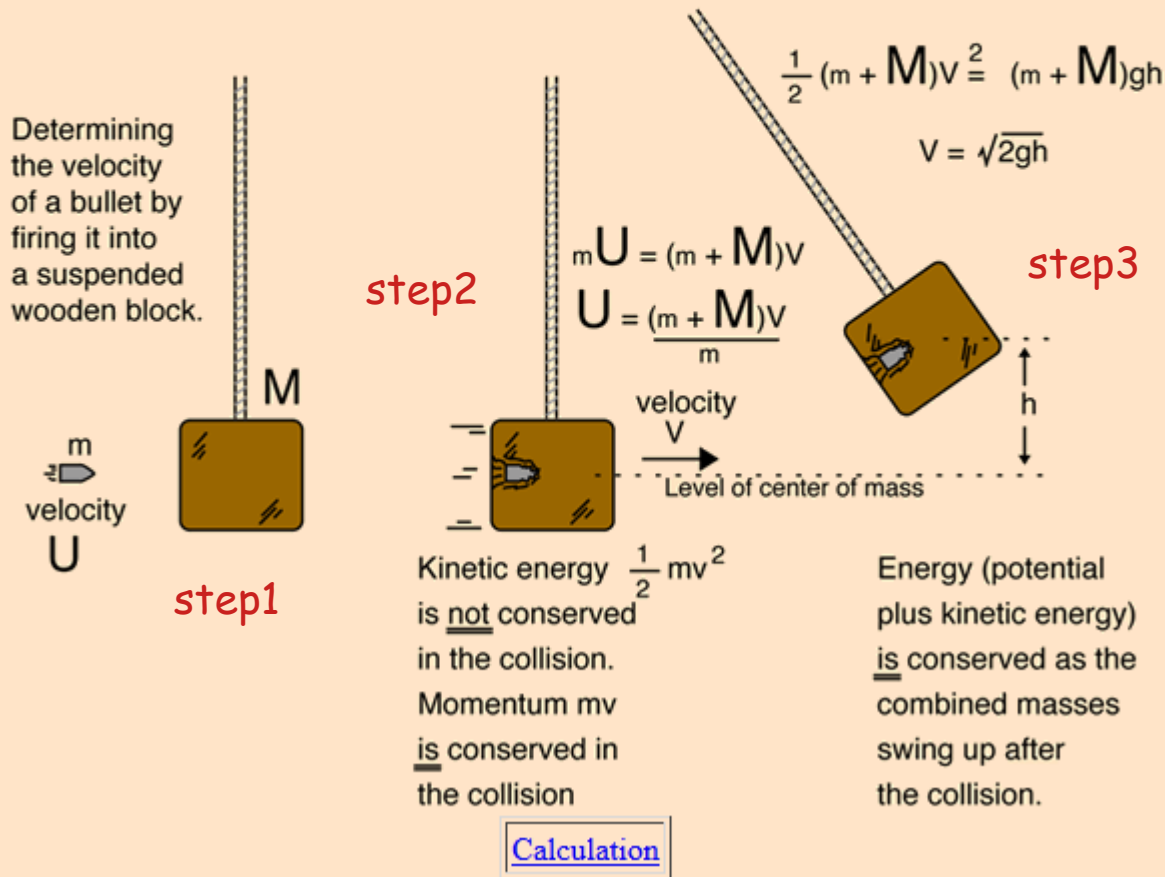
The 35-kilogram girl is standing on a 20-kilogram wagon and jumps off, giving the wagon a kick that sends it off at 3.8 meters per second. How fast is the girl moving?

A 1.2-kilogram basketball traveling at 7.5 meters per second hits the back of a 12-kilogram wagon and bounces off at 3.8 meters per second, sending the wagon off in the original direction of travel of the ball. How fast is the wagon going?

A 2 000-kilogram limousine going east at 22 meters per second strikes a 1 200-kilogram sports car going west at 30 meters per second. How fast are the two cars going together after the collision?

Ballistic Pendulum

The ballistic pendulum is a classic example of a dissipative collision in which [conservation of momentum](#) can be used for analysis, but [conservation of energy](#) during the collision cannot be invoked because the energy goes into inaccessible forms such as internal energy. After the collision, conservation of energy can be used in the swing of the combined masses upward, since the [gravitational potential energy](#) is [conservative](#).



→ The system is Block + bullet

→ we can't use conservation Of energy between step1 And step3 because we don't Know the energy lost To heat/noise

→ we can use conservation Of momentum between step1 And step2 because The frictional forces Are inside the system (bag With bullet and block) Gravity + tension Have no component along The horizontal (momentum is a vector)



7.3 Collisions in One Dimension

Example 8 A Ballistic Pendulum

The mass of the block of wood is 2.50-kg and the mass of the bullet is 0.0100-kg. The block swings to a maximum height of 0.650 m above the initial position.

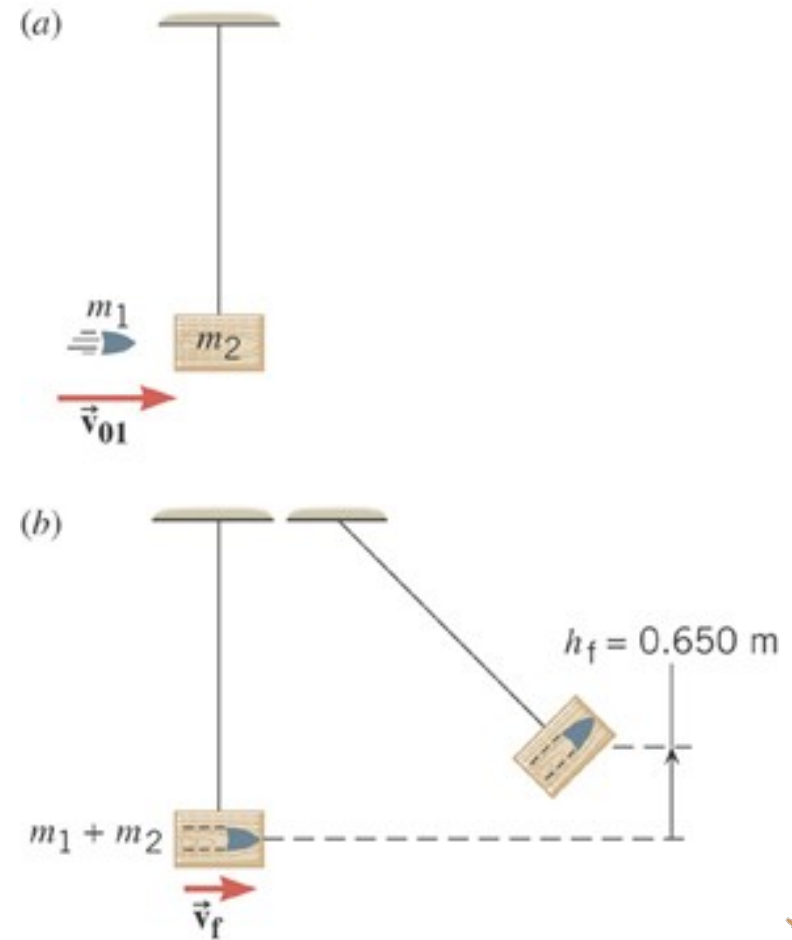
Find the initial speed of the bullet. → work backward.

Remember $PE = KE \rightarrow \text{speed} = \sqrt{2gh}$

Find speed of block+bullet

Then use conservation of momentum

Easy!



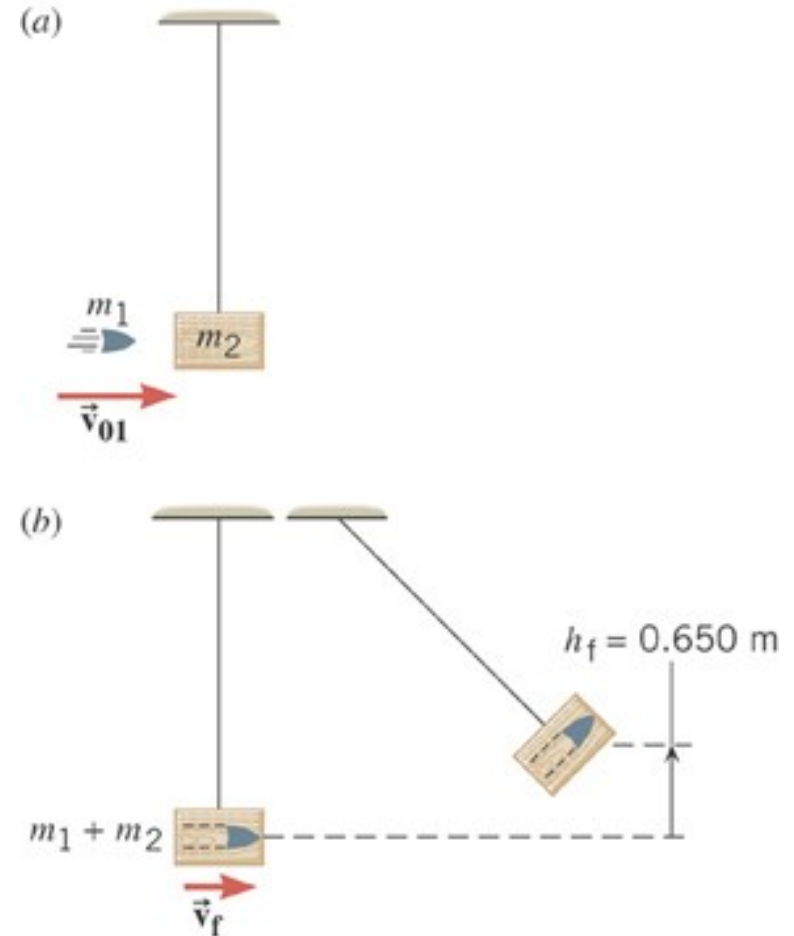
7.3 Collisions in One Dimension

Apply conservation of momentum to the collision:

$$m_1 v_{f1} + m_2 v_{f2} = m_1 v_{o1} + m_2 v_{o2}$$

$$(m_1 + m_2) v_f = m_1 v_{o1}$$

$$v_{o1} = \frac{(m_1 + m_2) v_f}{m_1}$$



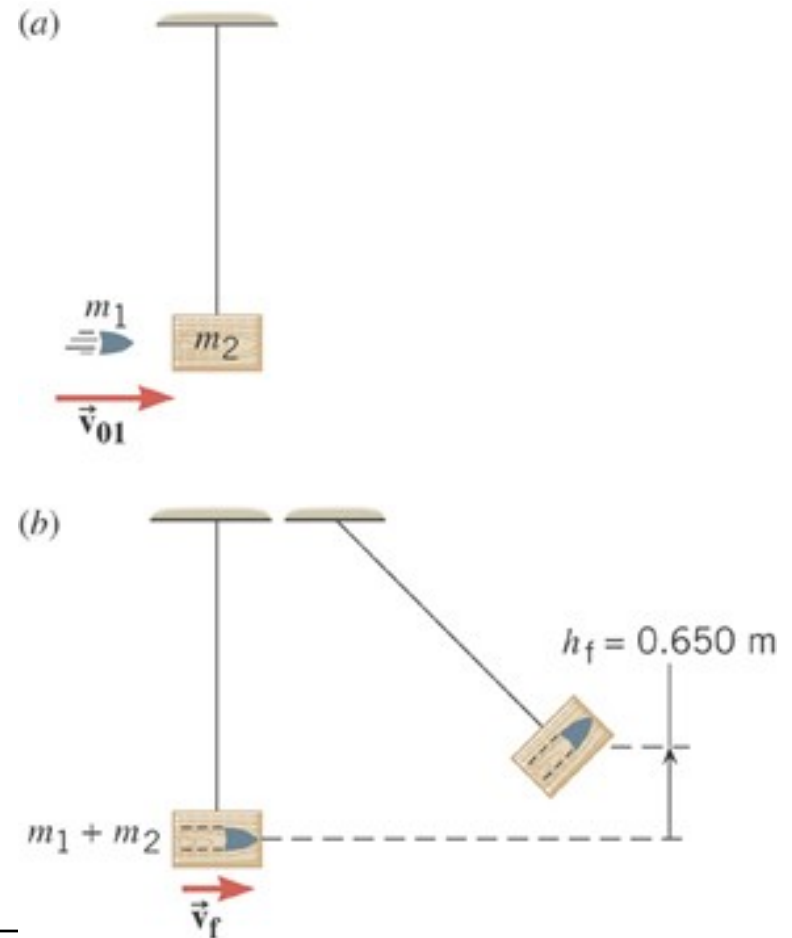
7.3 Collisions in One Dimension

$$mgh = \frac{1}{2}mv^2$$

$$(m_1 + m_2)gh_f = \frac{1}{2}(m_1 + m_2)v_f^2$$

$$gh_f = \frac{1}{2}v_f^2$$

$$v_f = \sqrt{2gh_f} = \sqrt{2(9.80 \text{ m/s}^2)(0.650 \text{ m})}$$

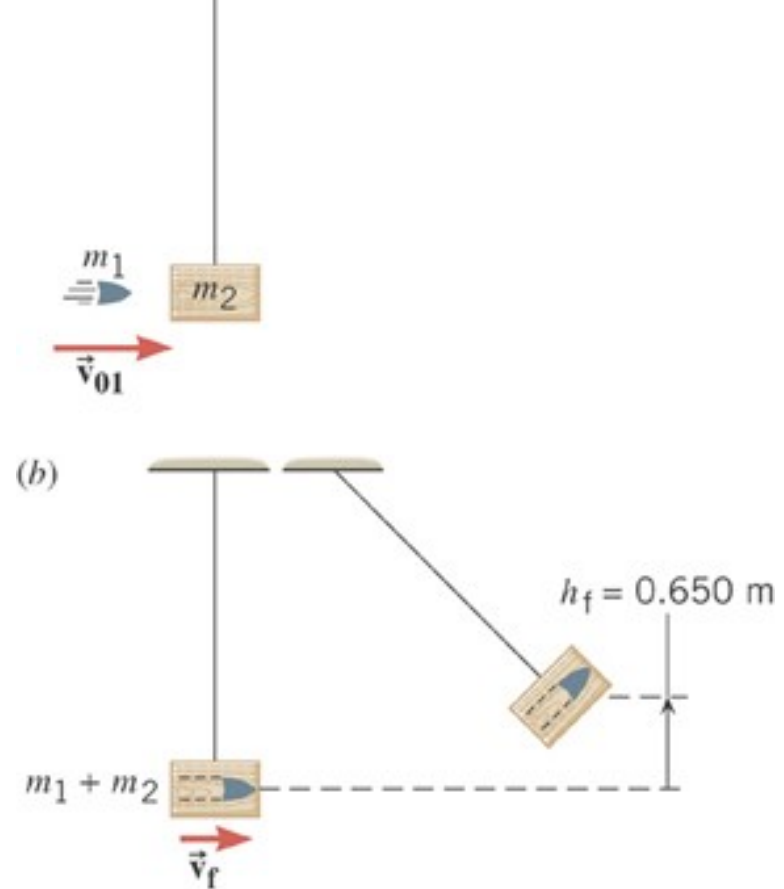


7.3 Collisions in One Dimension

$$v_f = \sqrt{2(9.80 \text{ m/s}^2)(0.650 \text{ m})}$$

$$v_{o1} = \frac{(m_1 + m_2)v_f}{m_1}$$

$$v_{o1} = \left(\frac{0.0100 \text{ kg} + 2.50 \text{ kg}}{0.0100 \text{ kg}} \right) \sqrt{2(9.80 \text{ m/s}^2)(0.650 \text{ m})} = +896 \text{ m/s}$$



When bullets fly through the air, they do so at amazing speeds. The fastest bullets travel more than 2,600 feet per second. That's equivalent to **over 1,800 miles per hour**. To put that in perspective, it's amazing to realize that bullets travel over twice the speed of sound!

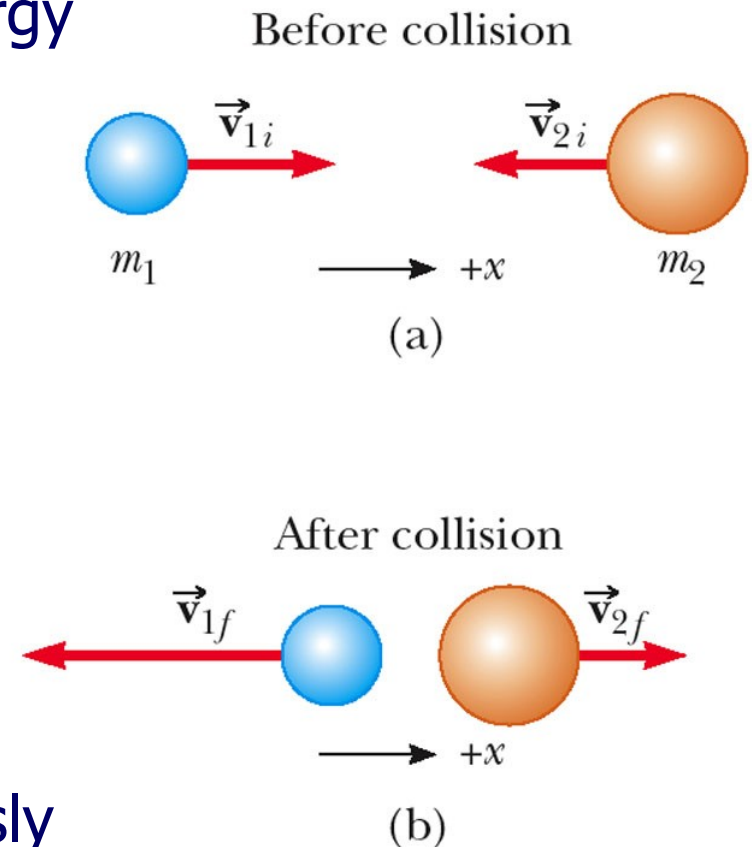


More About Elastic Collisions

- Both momentum and kinetic energy are conserved

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$
$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

- Typically have two unknowns
- Momentum is a vector quantity
 - Direction is important
 - Be sure to have the correct signs
- Solve the equations simultaneously



© 2006 Brooks/Cole - Thomson



Elastic Collisions

- A simpler equation can be used in place of the KE equation

$$\frac{1}{2}m_1v_{1i}^2 + \frac{1}{2}m_2v_{2i}^2 = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2$$

$$m_1(v_{1i}^2 - v_{1f}^2) = m_2(v_{2f}^2 - v_{2i}^2)$$

$$m_1(v_{1i} - v_{1f})(v_{1i} + v_{1f}) = m_2(v_{2f} - v_{2i})(v_{2f} + v_{2i})$$

$$m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f} \quad m_1(v_{1i} - v_{1f}) = m_2(v_{2f} - v_{2i})$$

basket ball
And ping
pong!

$$v_{1i} + v_{1f} = v_{2f} + v_{2i}$$

$$m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$$

$$v_{1i} - v_{2i} = -(v_{1f} - v_{2f})$$

Relative speed stays the same
Explain example of basket ball!

Problem Solving for 1D Collisions, 3

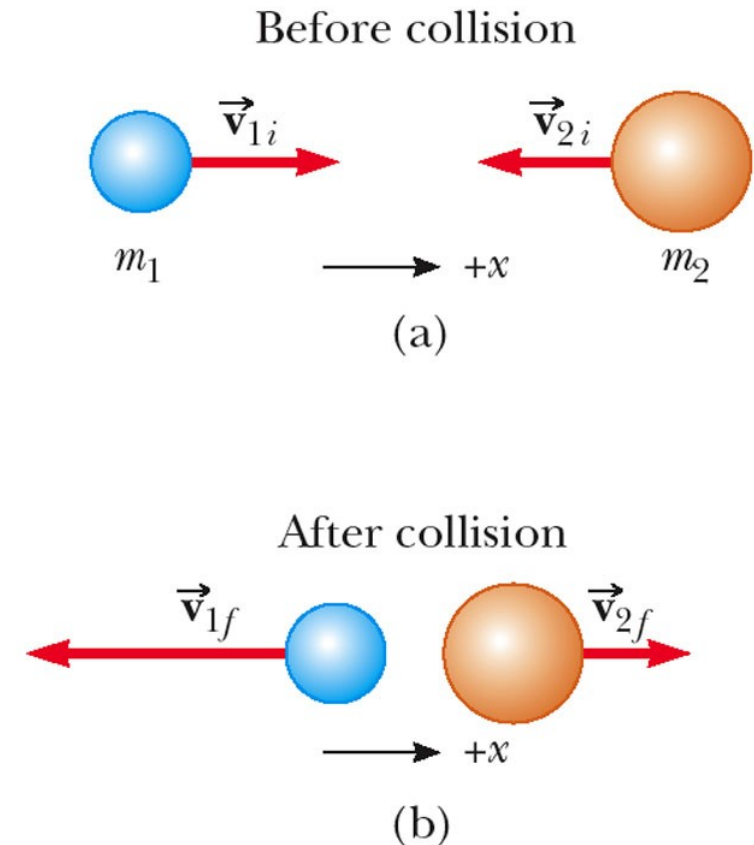
□ **Conservation of Energy:**

If the collision is elastic, write a second equation for conservation of KE, or the alternative equation

- This only applies to perfectly elastic collisions

$$v_{1i} + v_{1f} = v_{2f} + v_{2i}$$

□ **Solve:** the resulting equations simultaneously



© 2006 Brooks/Cole - Thomson



$$v'_2 = \frac{2m_1}{m_1 + m_2} v_1 - \frac{m_1 - m_2}{m_1 + m_2} v_2$$

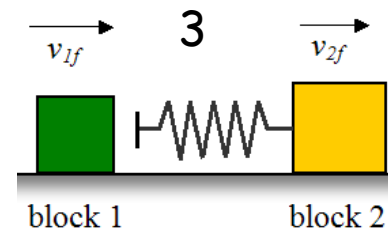
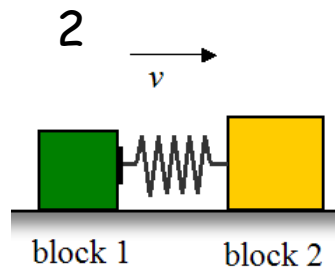
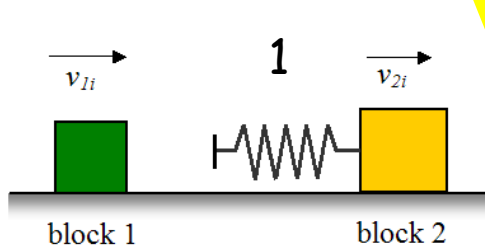
$$v'_1 = \frac{m_1 - m_2}{m_1 + m_2} v_1 + \frac{2m_2}{m_1 + m_2} v_2$$

For basketball and pingpong
 $M_1 \gg m_2$

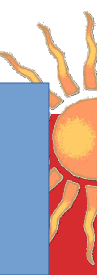
$V_1 = V$ (up) and $V_2 = -V$ (down)

$$V_2' = 2V - (-V) = 3V$$

$$V_1 = V_1$$



Consider 1 and 3. The collision is elastic. Same relative velocity.
 Because we neglect non conservative force. Spring force is conservative.
 Consider 1 and 2 → use conservation of energy. $KE_1 = PE_{el} + KE_2$

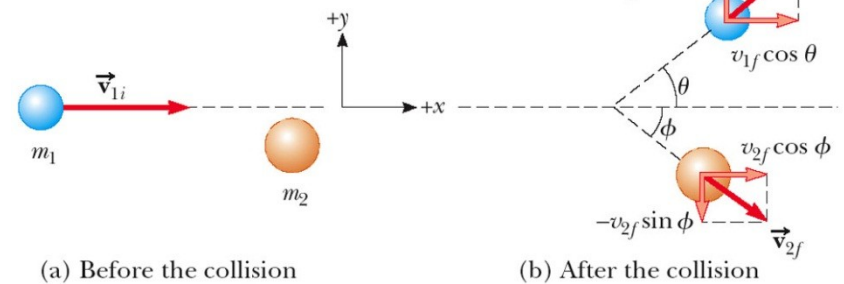


Two-Dimensional Collisions

- For a general collision of two objects in two-dimensional space, the conservation of momentum principle implies that the *total momentum of the system in each direction is conserved*

$$m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$

$$m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$



© 2006 Brooks/Cole - Thomson



Two-Dimensional Collisions

□ The momentum is conserved in all directions

□ Use subscripts for

$$m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$

■ Identifying the object

$$m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$

■ Indicating initial or final values

■ The velocity components

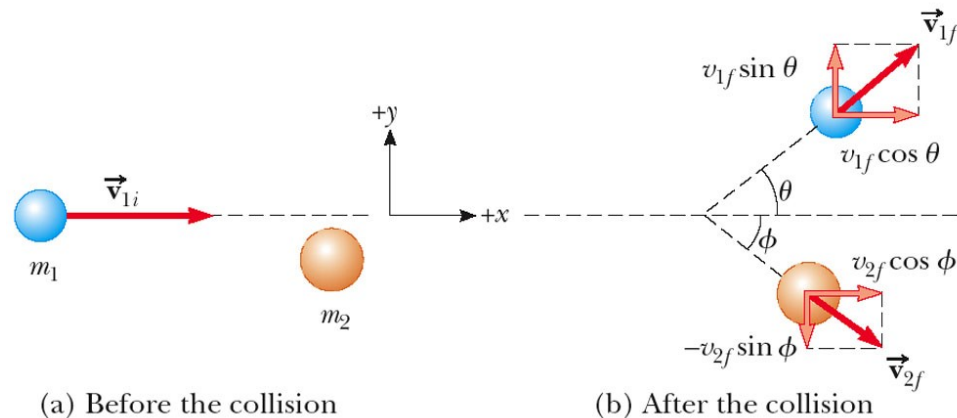
□ If the collision is elastic, use conservation of kinetic energy as a second equation

■ Remember, the simpler equation can only be used for one-dimensional situations

$$v_{1i} + v_{1f} = v_{2f} + v_{2i}$$



Glancing Collisions



© 2006 Brooks/Cole - Thomson

- ❑ The “after” velocities have x and y components
- ❑ Momentum is conserved in the x direction and in the y direction
- ❑ Apply conservation of momentum separately to each direction

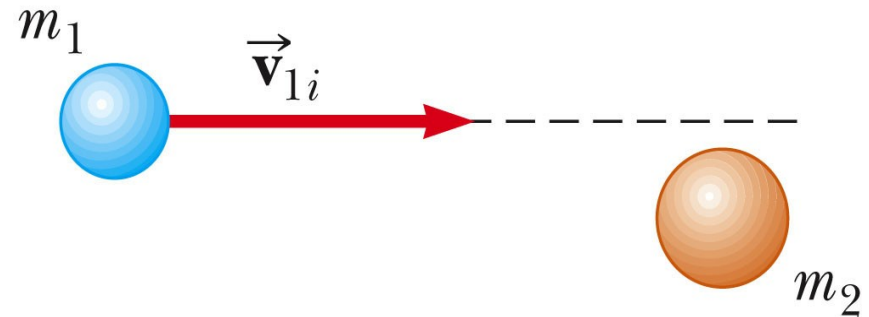
$$m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$

$$m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$



2-D Collision, example

- ❑ Particle 1 is moving at velocity $\vec{v}_{1i}$ and particle 2 is at rest
- ❑ In the x -direction, the initial momentum is $m_1 v_{1i}$
- ❑ In the y -direction, the initial momentum is 0



(a) Before the collision

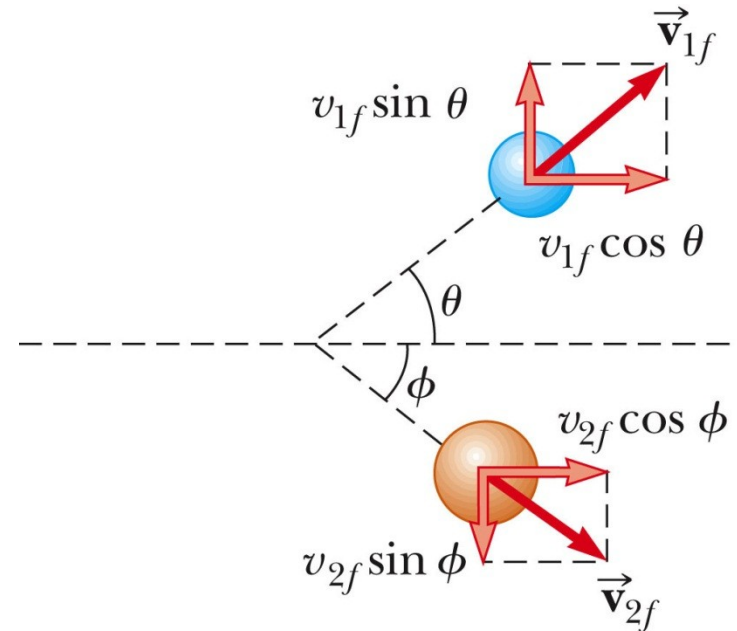
© 2007 Thomson Higher Education



2-D Collision, example cont

- After the collision, the momentum in the x -direction is $m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \phi$
- After the collision, the momentum in the y -direction is $m_1 v_{1f} \sin \theta + m_2 v_{2f} \sin \phi$

$$m_1 v_{1i} + 0 = m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \phi$$
$$0 + 0 = m_1 v_{1f} \sin \theta - m_2 v_{2f} \sin \phi$$



(b) After the collision

© 2007 Thomson Higher Education

- If the collision is elastic, apply the kinetic energy equation

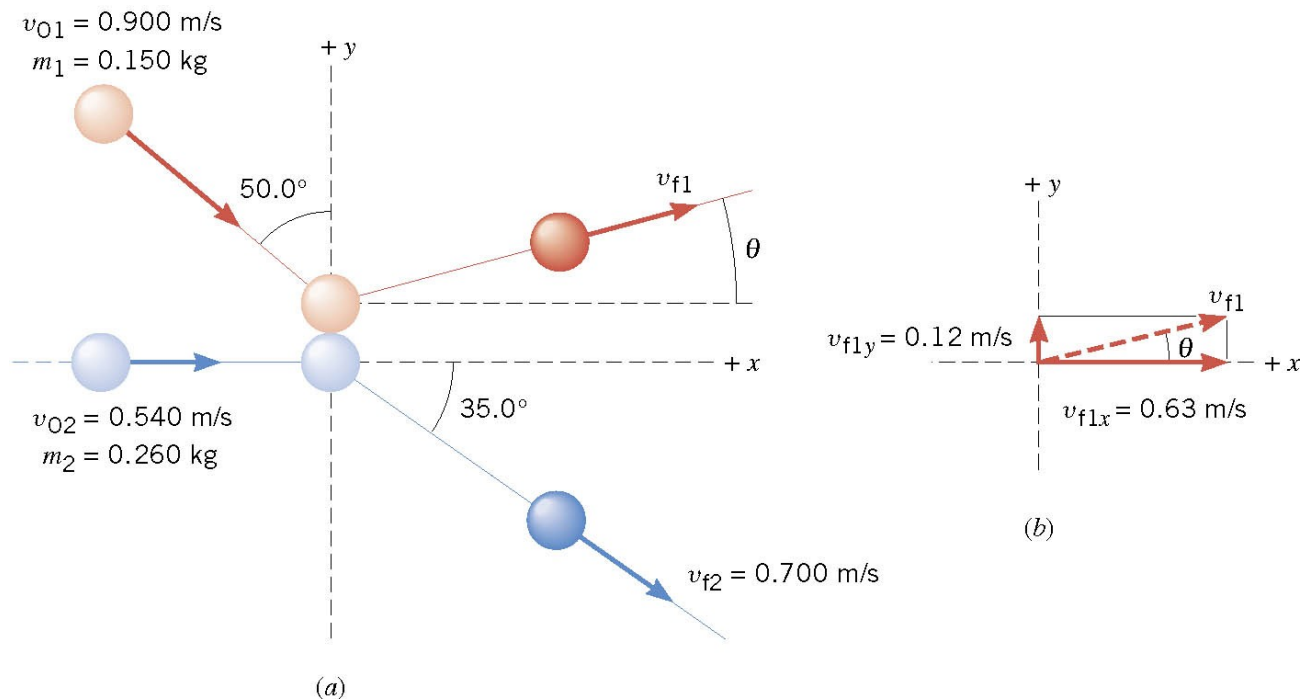
$$\frac{1}{2} m_1 v_{1i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$



7.4 Collisions in Two Dimensions

$$m_1 v_{f1x} + m_2 v_{f2x} = m_1 v_{o1x} + m_2 v_{o2x}$$

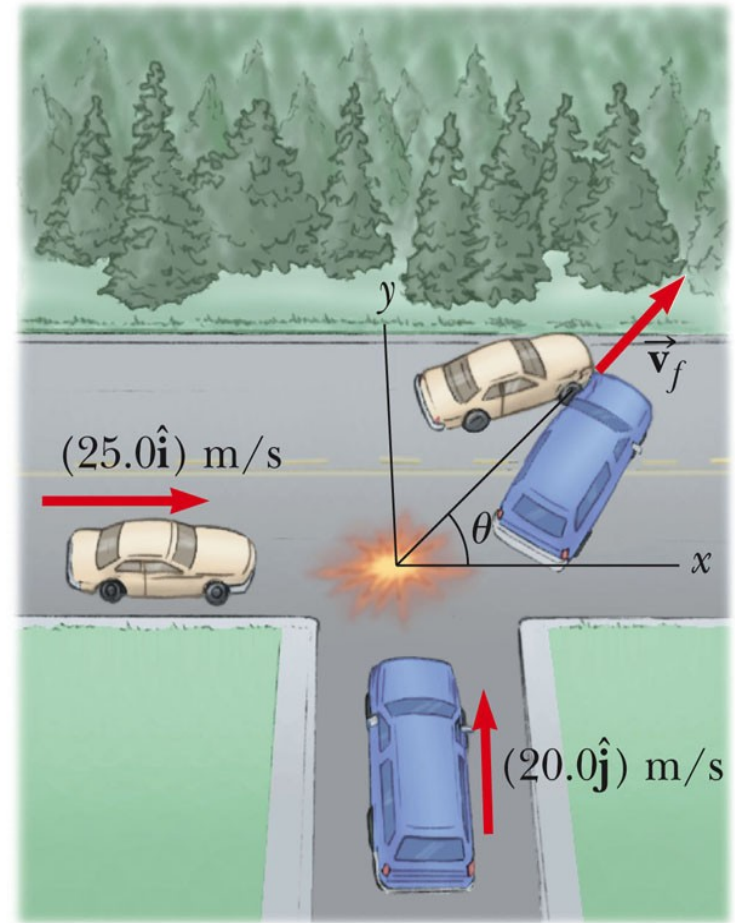
$$m_1 v_{f1y} + m_2 v_{f2y} = m_1 v_{o1y} + m_2 v_{o2y}$$



Collision at an Intersection

□ A car with mass $1.5 \times 10^3 \text{ kg}$ traveling east at a speed of 25 m/s collides at an intersection with a $2.5 \times 10^3 \text{ kg}$ van traveling north at a speed of 20 m/s . Find the magnitude and direction of the velocity of the wreckage after the collision, assuming that the vehicles undergo a perfectly inelastic collision and assuming that friction between the vehicles and the road can be neglected.

$$m_c = 1.5 \times 10^3 \text{ kg}, m_v = 2.5 \times 10^3 \text{ kg}$$
$$v_{cix} = 25 \text{ m/s}, v_{viy} = 20 \text{ m/s}, v_f = ? \theta = ?$$



© 2007 Thomson Higher Education



Collision at an Intersection

$$m_c = 1.5 \times 10^3 \text{ kg}, m_v = 2.5 \times 10^3 \text{ kg}$$

$$v_{cix} = 25 \text{ m/s}, v_{vix} = 20 \text{ m/s}, v_f = ? \theta = ?$$

$$\sum p_{xi} = m_c v_{cix} + m_v v_{vix} = m_c v_{cix} = 3.75 \times 10^4 \text{ kg} \cdot \text{m/s}$$

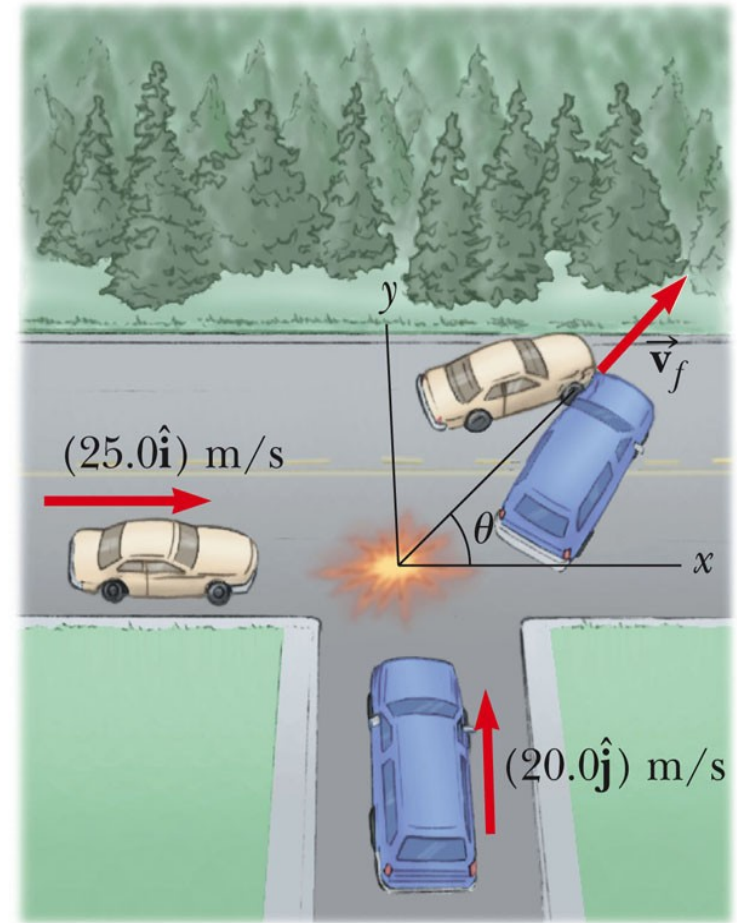
$$\sum p_{xf} = m_c v_{cfx} + m_v v_{vfx} = (m_c + m_v) v_f \cos \theta$$

$$3.75 \times 10^4 \text{ kg} \cdot \text{m/s} = (4.00 \times 10^3 \text{ kg}) v_f \cos \theta$$

$$\sum p_{yi} = m_c v_{ciy} + m_v v_{viy} = m_v v_{viy} = 5.00 \times 10^4 \text{ kg} \cdot \text{m/s}$$

$$\sum p_{yf} = m_c v_{cfy} + m_v v_{vfy} = (m_c + m_v) v_f \sin \theta$$

$$5.00 \times 10^4 \text{ kg} \cdot \text{m/s} = (4.00 \times 10^3 \text{ kg}) v_f \sin \theta$$



© 2007 Thomson Higher Education



Collision at an Intersection

$$m_c = 1.5 \times 10^3 \text{ kg}, m_v = 2.5 \times 10^3 \text{ kg}$$
$$v_{cix} = 25 \text{ m/s}, v_{viy} = 20 \text{ m/s}, v_f = ? \theta = ?$$

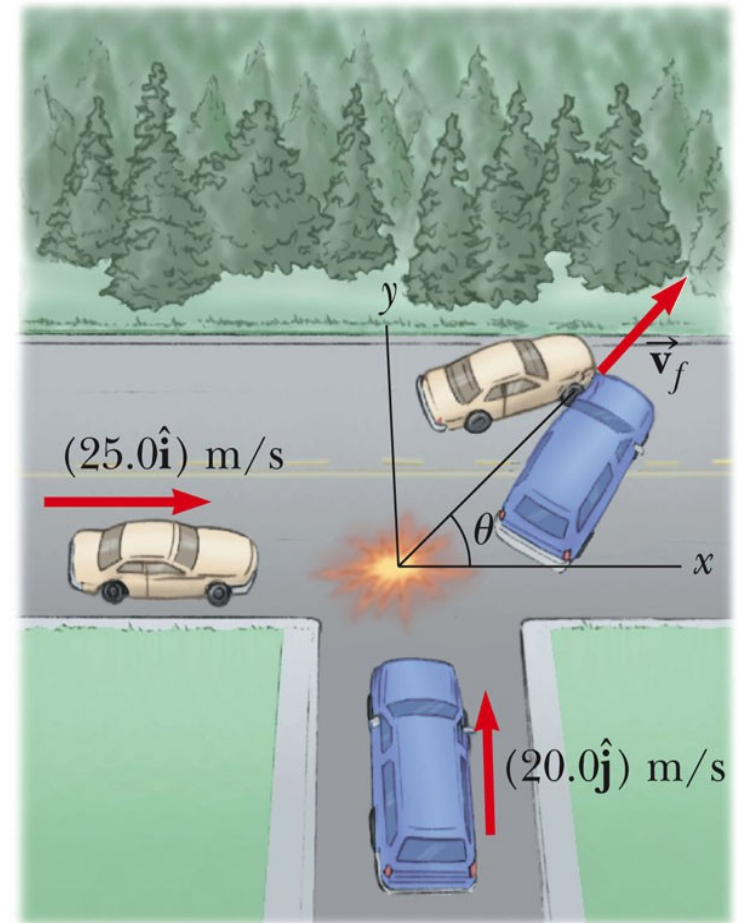
$$5.00 \times 10^4 \text{ kg} \cdot \text{m/s} = (4.00 \times 10^3 \text{ kg}) v_f \sin \theta$$

$$3.75 \times 10^4 \text{ kg} \cdot \text{m/s} = (4.00 \times 10^3 \text{ kg}) v_f \cos \theta$$

$$\tan \theta = \frac{5.00 \times 10^4 \text{ kg} \cdot \text{m/s}}{3.75 \times 10^4 \text{ kg} \cdot \text{m/s}} = 1.33$$

$$\theta = \tan^{-1}(1.33) = 53.1^\circ$$

$$v_f = \frac{5.00 \times 10^4 \text{ kg} \cdot \text{m/s}}{(4.00 \times 10^3 \text{ kg}) \sin 53.1^\circ} = 15.6 \text{ m/s}$$



© 2007 Thomson Higher Education

